“Predicting Future User Connections in Social Networks with Machine Learning Techniques”

A thesis report submitted to the department of Computer Science and Engineering of the World University of Bangladesh in partial fulfillment of the requirement for award of the Degree of Bachelor of Science in Computer Science & Engineering

Submitted by:

P. M. Niamul Hasan
Id: WUB 03/22/60/3563

Md. Ahsan Habib Sumon
Id: WUB 03/22/60/3566

Anup Kumar Dey
Id: WUB 03/22/60/3569

Supervised by:

MD. Mahedi Hassan, Lecturer
Dept. of Computer Science and Engineering



World University of Bangladesh

Department of Computer Science and Engineering

July, 2025

LETTER OF TRANSMITTAL

July 5, 2025

To
MD. Mahedi Hassan
Lecturer, Department of Computer Science & Engineering
World University of Bangladesh
(WUB) Avenue 6, Lake Drive Road,
Sector: 17/H, Uttara, Dhaka - 1230

Subject: Submission of Project Report.

Dear Sir,

We are pleased to submit the thesis report entitled **" Predicting Future User Connections in Social Networks with Machine Learning Techniques "**. It was a great pleasure to work on such an important topic. The thesis report is prepared according to the requirements and guidelines of the Department of Computer Science and Engineering, World University of Bangladesh (WUB).

We believe that the thesis report will help you in evaluating our thesis work. It would be a great pleasure for us to interpret any part or whole of the report whenever necessary.

Sincerely yours:	Sincerely yours:	Sincerely yours:
---	---	---
P. M. Niamul Hasan	Md. Ahsan Habib Sumon	Anup Kumar Dey
Id: WUB 03/22/60/3563	Id: WUB 03/22/60/3566	Id: WUB 03/22/60/3569



World University of Bangladesh

DECLARATION

We hereby solemnly declare that the thesis work entitled "**Predicting Future User Connections in Social Networks with Machine Learning Techniques**", has been supervised by MD. Mahedi Hassan, Lecturer, Department of Computer Science & Engineering, World University of Bangladesh. We ensure that the thesis report has not been submitted either in whole or part for any degree or Diploma in any university previously.

We hereby warrant that the work we have presented does not breach any existing copyright rule.

We further undertake to indemnify the university against any loss or damage arising from breach of the forgoing obligation.

Sincerely yours:	Sincerely yours:	Sincerely yours:
---	---	---
P. M. Niamul Hasan	Md. Ahsan Habib Sumon	Anup Kumar Dey
Id: WUB 03/22/60/3563	Id: WUB 03/22/60/3566	Id: WUB 03/22/60/3569



Department of Computer Science and Engineering

World University of Bangladesh

CERTIFICATE

We hereby certify that the Thesis Report on "**Predicting Future User Connections in Social Networks with Machine Learning Techniques**", is a confide record of thesis work done by P. M. Niamul Hasan, Md.Ahsan Habib Sumon , Anup Kumar Dey for partial fulfillment of the requirements for award of the degree of the Bachelor of Science in Computer Science and Engineering from World University of Bangladesh.

The thesis report has been carried out under my guidance and is a record of the bona-fide work carried out successfully by the students.

Supervisor:

……

MD. Mahedi Hassan

Lecturer of the department of Computer Science and Engineering

World University of Bangladesh (WUB)

ACKNOWLEDGEMENTS

We extended our heartfelt gratitude to almighty GOD for supporting us throughout our thesis journey. Without His blessing, this accomplishment would not have been possible.

Our sincere appreciation goes to our thesis supervisor, **Md. Mahedi Hassan,** Lecturer of the department of computer science and engineering at WUB. His invaluable guidance, patience, enthusiasm, co-operation and insightful ideas were instrumental in shaping the research on **" Predicting Future User Connections in Social Networks with Machine Learning Techniques".** His constant support motivated me to complete this work in its present form.

We also wish to express our special thanks to all the teachers in the Department of Computer Science and Engineering at World University of Bangladesh. Their unwavering support and cooperation played a significant role in our academic journey and enabled us to successfully.

Lastly, we are deeply grateful to our parents, whose love and care have been a constant source of strength and inspiration. Their sacrifice and encouragement have been invaluable.

To everyone, who played a part, big or small, in this thesis and our academic journey, we offer sincere thanks. Your contributions have been vital to our success, and we are truly grateful for your presence in our life.

Sincerely yours:	Sincerely yours:	Sincerely yours:
---	---	---
P. M. Niamul Hasan	Md. Ahsan Habib Sumon	Anup Kumar Dey
Id: WUB 03/22/60/3563	Id: WUB 03/22/60/3566	Id: WUB 03/22/60/3569

ABSTRACT

This thesis focuses on the prediction of future user connections in social networks through the application of various machine learning techniques. By combining both topological features (node degree, common neighbors) and non-topological features (user behavior, preferences) the study aims to enhance the accuracy of predicting future links between users in dynamic social networks. The dataset used in this research includes features from Facebook with both structural and user interaction data. Our dataset consists of 21 features and 2997 instances which offer a balanced representation of user connections and behaviors thus enabling a more nuanced analysis of prediction patterns.

The study explores a wide range of machine learning algorithms including Random Forest, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Ensemble models and Deep Learning approaches like Keras DNN to predict future user connections. These models leverage both the inherent network topology and behavioral patterns of users to improve prediction accuracy. The performance of these models is evaluated using standard metrics such as accuracy, precision, recall, F1-score and AUC with the results indicating that the Keras DNN model outperforms other models with an AUC of 0.9863 and a high accuracy of 92.8%. Additionally, ensemble methods like AdaBoost and Gradient Boosting contribute to improving the reliability and robustness of predictions by mitigating biases and overfitting.

Furthermore, the study incorporates Explainable Artificial Intelligence (XAI) techniques, particularly SHAP (SHapley Additive exPlanations) to provide transparency and interpretability in model predictions. This approach allows us to better understand the contribution of various features in predicting future links offering valuable insights for practical applications. The findings emphasize the importance of both network structure and user-specific behavior in making robust and reliable predictions. These insights have significant potential for applications such as enhancing friend recommendation systems, detecting potential threats or fraud and optimizing user engagement strategies in e-commerce and social media platforms.

Keywords: Link prediction, Social networks, SMOTE, Machine learning, Deep learning, Keras DNN, XAI, SHAP, Facebook dataset.

Table of contents **Page**

Letter of Transmittal ……..... ii

Declaration ……..... iii

Certificate……..…….. iv

Acknowledgement ……..... v

Abstract …….. vi

Table of Contents …….. vi

Chapter: 1 Introduction …….. **1-4**

1.1. Introduction……..... 1

1.2 Objectives …….…….... 2

1.3 Justification of Study ……. 2

1.4 Scopes of Study ……. 3

Chapter: 2 Literature Review…….... 5-7

2.1 Review of the literature …… 5

Chapter: 3 Methodology…… 8-14

3.1 Methodology Overview……. 8

3.2 Justification of methodology ……. 9

3.3 Description of Methodology ……. 10

3.3.1 Dataset Description …….. 10

3.3.2 Data Preprocessing …….. 13

Chapter: 4 Requirement Analysis ……..…….. 15-28

4.1 Dataset Collection ……. 15

4.2 ML Model Algorithms Used in This Analysis ……. 15

4.2.1 Random Forest …… 15

4.2.2 Support Vector Machines (SVM) …… 15

4.2.3 k-Nearest Neighbors (k-NN) …… 16

4.2.4 XGBoost ……..... 17

4.2.5 Voting Ensemble ……..... 18

4.2.6 Stacking Ensemble …….. 18

4.2.7 Bagging Ensemble …… 19

4.2.8 AdaBoost Ensemble …….. 20
4.2.9 Gradient Boosting Ensemble ……. 21
4.2.10 Keras DNN …….. 22
4.2.11 Transformer ……..... 23
4.2.12 MLP Classifier ……. 24
4.3 Performance Measurement Techniques ……. 25
Chapter: 5 Result and Analysis ……..…….. 29-62
5.1 Exploratory Data Analysis (EDA)……..... 29
5.2 Performance of the ML Algorithms…… 33
5.2.1 Performance Summary …… 34
5.3 Result of the XAI Tools ……. 47
5.3.1 keras DNN Global Explainability …… 47
5.3.2 Local Explainability ……. 51
5.4 Comparison of k-NN Model with XAI Tools ……..... 54
5.4.1 KNN Global Explainability ……. 54
5.4.2 Local Explainability …… 58
5.5 XAI (Explainable AI) Results Comparison Summary …….. 62
Chapter: 6 Conclusions …… 63-67
6.1 Conclusions ……... 63
6.2 Limitations & Recommendation…..……..…… 64
6.3 Future works…….. 64
References……..... **64**
Formula References …….. 66

List of tables

Table No.	Title	page
Table 3.3.1	Example of Features in the Dataset	11
Table 4.3	Performance Evaluation Metric	25
Table 5.2.1	ML Models Performance Comparison	34
Table 5.2.2	Ensemble Models Performance Comparison	35
Table 5.2.3	Deep Learning Models Performance Comparison	37
Table 5.2.4	All Model Performance Comparison Predicting Future User Connections in Social Networks	38
Table 5.2.5	Keras DNN Scoring Report	43
Table 5.2.6	K-Fold Cross Validation Performance Configuration for KerasDNN	44
Table 5.3.1	Global Feature Importance using SHAP (Gradient Explainer)	49
Table 5.3.9	Global Feature Importance using SHAP (Kernel Explainer)	56

List of Figures

Figure No.	Title	page
Figure 3.1	Workflow of the proposed Methodology	8
Figure 5.1.1	Analyzing of Facebook Source Distribution and Boxplot	29
Figure 5.1.2	Analyzing of Facebook Target Distribution and Boxplot	29
Figure 5.1.3	Correlation Analysis of Source and Target Features	30
Figure 5.1.4	Correlation Pair Plot Analysis of Source and Target Features	31
Figure 5.1.5	Analyzing Correlation Matrix Across User, Interaction, and Network Datasets	32
Figure 5.2.1	ML Models Performance Comparison Barchart	34
Figure 5.2.2	Ensemble Models Performance Comparison Barchart	36
Figure 5.2.3	Deep Learning Models Performance Comparison Barchart	37
Figure 5.2.4	Deep Learning Models Performance Comparison Barchart	39
Figure 5.2.5	ROC Curves for Multiple Classifiers in Link Prediction for Facebook's Future User Connections	40
Figure 5.2.6	Confusion Matrix for Keras DNN Classifier	41
Figure 5.2.7	Precision-Recall Curve for Keras DNN	42
Figure 5.2.8	K-Fold Cross Validation Configuration for Keras DNN	44
Figure 5.2.9	CSV Dataset Network Graph	45
Figure 5.2.10	Facebook Dataset Network Graph	46
Figure 5.2.11	Combined Network Graph	46
Figure 5.3.1	Importance of global features of the Best keras DNN Model	48
Figure 5.3.2	Mean SHAP Values for keras DNN Model	50
Figure 5.3.3	Contribution of the feature "Interests" In the model	51
Figure 5.3.4	Contribution of the feature "Gender" In the model	52
Figure 5.3.5	Contribution of the feature " Degree_u " and " Shortest Path " In the model	52
Figure 5.3.6	Contribution of the feature " shortest path " In the model	53
Figure 5.3.7	Contribution of the feature " same community " In the model	53
Figure 5.3.8	Contribution of the feature " u Interests " In the model	53

Figure 5.4.1	Importance of global features of the Best K-NN Model	55
Figure 5.4.2	Mean SHAP Values for k-NN Model	57
Figure 5.4.3	Contribution of the feature " Shortest Path" In the model	58
Figure 5.4.4	Contribution of the feature "Location" and "Activity Level" In the model	58
Figure 5.4.5	Contribution of the feature "Shortest Path" and "Same Community" In the model	59
Figure 5.4.6	Contribution of the feature " shortest path " In the model	60
Figure 5.4.7	Contribution of the feature "Same Community" In the model	60
Figure 5.4.8	Contribution of the feature " Preferential Attachment " In the model	61

Abbreviations

Short Form	Full Form
AI	Artificial Intelligence
ML	Machine Learning
K-NN	k-Nearest Neighbors
SVM	Support Vector Machine
XAI	Explainable artificial intelligence
EM	Ensemble Models
SHAP	SHapley Additive exPlanations
MLP	Multilayer Perceptron

Chapter 1: Introduction

1.1 Introduction

In the modern digital world social networks play a major role in how people connect, communicate and share information. Social networks such as Facebook, Instagram, Twitter and LinkedIn are made up of individuals (also called nodes) and the connections (or edges) between them. These networks grow and evolve over time with people constantly making new friends, finding business contacts or building new relationships. The task of Predicting Future User Connections is about figuring out which new connections are likely to happen in the future based on the current structure of the network. In other words, it is the process of predicting which people (or nodes) will connect with each other in the future.

Predicting Future User Connections is a very important and exciting problem as it has applications in many areas of life. In social networks Predicting Future User Connections can be used to recommend new friends, business partners or professional contacts making the experience of using these networks much more personalized. For example Facebook recommends new friends based on mutual friends or common interests. In security predicting future links can help detect and stop criminal or terrorist activity by identifying possible connections between individuals before they form. Predicting Future User Connections also plays a significant role in recommendation systems (for example suggesting new products, services or media content) where systems can predict user preferences and recommend products, movies or items based on their predicted interests.

The goal of this study is to improve our understanding of Predicting Future User Connections and make it more accurate by using machine learning techniques**.** Machine learning is a field of artificial intelligence (AI) that helps computers learn patterns from data and make decisions without being explicitly programmed. By using machine learning, we can better predict which new links will form in a network.

1.2 Objectives

The objectives of this research are as follows:

1. To create a comprehensive dataset that includes both traditional network features (topological) and non-topological features (such as user behavior, preferences and external factors) enabling more robust predictions.

2. To combine topological and non-topological features for more accurate Predicting Future User Connections improving traditional methods that rely solely on network structure.

3. To explore and apply machine learning algorithms such as Random Forest, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Ensemble Models and Deep Learning approaches to improve the accuracy of Predicting Future User Connections.

4. To incorporate Explainable Artificial Intelligence (XAI) techniques in order to provide interpretability and transparency in the prediction process allowing a better understanding of how and why specific Predicting Future User Connections are made.

1.3 Justification of Study

The reason why this study is important is that it combines two types of features topological (network structure) and non-topological (user behavior, preferences etc.) to make more accurate predictions. While traditional methods focused only on topological features. This research takes into account additional information such as what people do in the network, their interactions and their preferences. This will allow us to better understand how networks evolve and predict future links with more accuracy. Additionally, this study will compare several machine learning models to determine which one works best for Predicting Future User Connections . The research will includes both topological and non-topological features making the Predicting Future User Connections model more comprehensive and robust. By combining the two types of features the research will show how we can improve the accuracy of predictions by using machine learning.

The practical applications of this research are vast. In social networks better Predicting Future User Connections can lead to more personalized recommendations helping platforms suggest friends, groups or events that users are likely to be interested in. For security, predicting new links between people can help identify potential threats and prevent criminal or terrorist activities. In e-commerce better recommendation systems can suggest products or services based on predicted user interests, increasing sales and customer satisfaction. As social networks and other online networks continue to grow the ability to predict new links will become even more important. This research will contribute to the growing field of Predicting Future User Connections by showing how combining different features (topological and non-topological) with machine learning can improve predictions. The results of this study can be applied in a wide range of areas including social media, security, e-commerce and recommendation systems making it a valuable contribution to the understanding of how relationships form and evolve in networks.

1.4 Scope of Study

The scope of this study is to explore and enhance the field of Predicting Future User Connections in social networks using machine learning techniques. The research will focus on improving the accuracy of predictions by integrating both topological (network structure) and non-topological features (user behavior, preferences, etc.). By applying machine learning models such as Random Forest, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Ensemble Models and Deep Learning approaches this study will investigate how these techniques can be utilized to predict future connections in social networks.

The study will also create a comprehensive dataset that includes traditional network features alongside user-specific behavioral data providing a more holistic view of Predicting Future User Connections. The research will primarily be applied to social networks like Facebook, Instagram, and LinkedIn but the techniques developed could be generalized to other network types including security systems, e-commerce and recommendation systems. This study is not intended to create an exhaustive framework for all Predicting Future User Connections scenarios but will focus on comparing different machine learning models and determining which offers the most accurate predictions for the types of networks considered.

Additionally, the research will explore how various machine learning models can handle different network sizes and user behaviors providing a comparison of performance across various data scales. The potential for scalability will be assessed ensuring that the proposed techniques can be applied to both small and large-scale social networks. Furthermore, ethical considerations such as user privacy and data security, will be addressed by ensuring the dataset complies with data protection standards. The findings will provide practical insights for improving social network recommendations identifying potential security threats and enhancing product recommendations in commercial platforms. This study aims to create a benchmark for future work in the field and offer new perspectives on using machine learning in social network prediction tasks.

Chapter 2: Literature Review

2.1 Review of the Literature

Link prediction in social networks focuses on forecasting future or missing connections between users. It has broad applications across domains such as social media friend suggestions, collaborative filtering in recommendation systems and enhancing system security by detecting anomalous links.

Traditional approaches primarily relied on topological features including node degree, common neighbors or Jaccard similarity. While effective to some extent these models often overlooked non-topological features such as user demographics or historical interactions factors that are critical in real-world prediction contexts.

To address this gap recent studies have emphasized integrating non-topological data including demographic information (id, age, location, interests, activity etc.) communication history and behavioral attributes. These additional dimensions enable models to better reflect the multifaceted and dynamic nature of real social interactions. Incorporating such contextual features leads to more nuanced and accurate predictions.

Contextual features were combined with traditional topological metrics to improve prediction performance. By integrating user-level data such as demographics alongside network structure models achieved better accuracy. This combination provided richer insights into potential link formation patterns. (Lichtenwalter, Lussier and Chawla, 2010)

Even basic heuristics like Common Neighbors showed improved results when enriched with contextual information such as shared interests or topical similarity demonstrating the value of incorporating user context. This highlights the strength of simple metrics when paired with relevant behavioral data. (Z. Liu, Q. M. Zhang, L.L u and T. Zhou, 2011)

Ensemble learning techniques outperformed logistic regression by leveraging both topological and behavioral features. Random Forest and boosting methods achieved significantly higher AUC

scores showing the strength of model diversity. The inclusion of varied features enhanced the model's robustness. (Anuradha Annaswamy, Henry Samueli and Boxue Du, 2014)

Deep learning models have emerged as powerful tools for link prediction in social networks. They effectively capture complex patterns and relationships in large, dynamic datasets. These models outperform traditional methods by learning deep node representations and adapting to evolving structures. The study shows improved accuracy in predicting future user connections using neural networks. It highlights the strength of deep learning in handling real-world, large-scale networks. (Muhan Zhang and Yixin Chen, 2018)

Supervised learning methods were shown to be effective for high-dimensional social datasets offering both scalability and flexibility. These models handled complex feature interactions efficiently in large networks. Their performance in managing diverse data types made them suitable for real-time systems. (Mohammad G. Raeini, 2020)

This review highlights the diverse applications of link prediction in social networks, such as friend recommendations, expert finding and fraud detection. It categorizes methods into supervised and unsupervised approaches based on data context. The paper emphasizes the growing use of contextual and behavioral features alongside topological metrics. Applications span beyond social platforms to areas like e-commerce and cybersecurity. Key challenges include data sparsity, scalability and dynamic network changes. The authors provide a structured overview of techniques and their practical impact in various domains.(Nur Nasuha Daud, Siti Hafizah ab Hamid, Muntadher Saadoon, Firdaus Sahran and Nor Badrul Anuar, 2020)

This study proposes a link prediction method for multiplex social networks using local random walks. The approach focuses on identifying dependable pathways across multiple network layers. By leveraging local structural information it improves prediction accuracy without heavy computational cost. The method is effective in handling the complexity of multilayer relationships in social platforms. (Wenjun Li, Ting Li and Kamal Berahmand, 2022)

Temporal patterns and multimodal signals were integrated into context-aware architectures to model evolving social behavior more accurately. This approach emphasized the importance of

dynamic and diverse data types. It allowed models to adapt over time to shifting user behaviors. (Meng Qin, Dit Yan Yeung, 2023)

A generative GNN based framework, GraphLP learned latent node representations using probabilistic encoding. This method effectively modeled both temporal dynamics and complex dependencies within networks. Tested across multiple social datasets it achieved high AUC scores between 0.92 and 0.95 demonstrating strong performance in dynamic environments. (Xing ping Xian, Tao Wu, Xiaoke Ma, Shaojie Qiao, Yabin Shao, Chao Wang, Lin Yuan and Yu Wu, 2023)

An ensemble-based approach was developed using Random Forest, XGBoost and rich feature extraction techniques for link prediction in directed social networks. Evaluated on a Facebook dataset the model successfully integrated topological, behavioral and directional attributes. It achieved an AUC of 0.883 showing the power of combining multiple algorithms for improved accuracy.(Sheroz Khan, Mohamed Badiy, Fatima Amounas and Mourade Azrour, 2024)

Real-world social platforms like Facebook, Twitter and LinkedIn employ link prediction systems for friend suggestions, fraud detection and growing user engagement. These systems utilize a hybrid of structural, behavioral and contextual data to suggest relevant and trustworthy connections. The integration of diverse features ensures more accurate and personalized recommendations. (Xiaoli Huang, Jingyu Lit and Yumiao Yuan, 2024)

Structural uncertainty in dynamic networks was addressed using probabilistic models like dynamic Bayesian networks which modeled temporal evolution and unpredictable changes in social links. These models improved reliability in fast-changing social environments. (Mengdi Sun, Minghu Tang, 2025)

Chapter 3: Methodology

3.1 Methodology Overview

This section outlines the methodology for improving the accuracy of machine learning models in Predicting Future User Connections within social networks. The methodology involves several key stages, including data collection, data preprocessing and the use of machine learning algorithms for Predicting Future User Connections. The workflow as depicted in Figure 3.1.

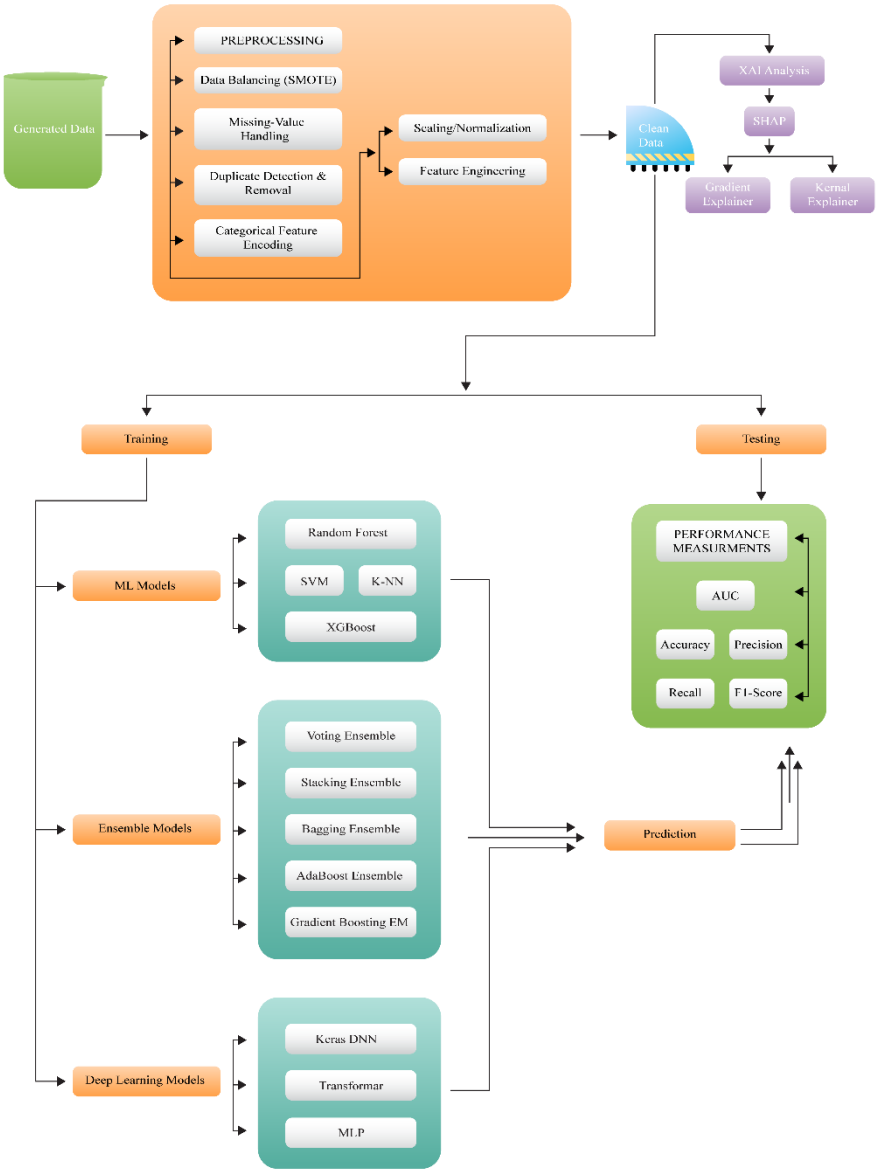


Figure 3.1: Workflow of the proposed Methodology

In Figure 3.1 The methodology for this research employs a hybrid approach that integrates both topological and non-topological features to enhance the accuracy of Predicting Future User Connections in social networks. Topological features refer to the structural aspects of the network. Non-topological features on the other hand pertain to user behavior, preferences and interaction history which offer deeper insights into how users engage with the network beyond mere connections. To analyze this data we use several machine learning algorithms including Random Forest, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Ensemble Models and Deep Learning approaches. These algorithms have been selected for their proven effectiveness in handling complex high-dimensional data and their ability to model intricate relationships in large datasets. The comprehensive dataset which includes both network structure and user behavior data provides a more holistic view of the interactions within the social network allowing for more accurate Predicting Future User Connections.

To evaluate the effectiveness of the different machine learning models we will use a variety of performance metrics including accuracy, precision, recall and F1 score. These metrics will allow us to assess how well each model is performing and to make comparisons between different approaches. By conducting a comparative analysis of these algorithms we will determine which machine learning model provides the most reliable and effective predictions for Predicting Future User Connections tasks. The workflow from raw data collection to the final prediction step has been carefully designed to ensure that each process is streamlined for optimal prediction accuracy. This stepwise approach minimizes errors ensures the correct handling of features and improves the overall robustness of the model. Furthermore, by focusing on the combination of topological and non-topological features the methodology aims to provide a more comprehensive and accurate solution to the Predicting Future User Connections problem in social networks.

3.2 Justification of Methodology

The chosen methodology combines multiple machine learning models to maximize the accuracy of Predicting Future User Connections . We have selected classifiers such as Random Forest, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Ensemble Models and Deep Learning approaches. The justification for this approach stems from their ability to handle large datasets and their proven effectiveness across various domains including social network

analysis and fraud detection. By integrating both topological and non-topological features our model aims to predict the likelihood of future connections based on existing interactions. The use of ML algorithms, Ensemble Models and Deep Learning approaches allows the model to capture intricate relationships within complex datasets making it well-suited for Predicting Future User Connections in dynamic social networks. These models can learn from both structural network features (node degree, common neighbors and clustering coefficients) and user-specific behaviors (interaction history and preferences). By using this hybrid approach the model can predict connections with more accuracy accounting for not just the network structure but also behavioral patterns of the individuals involved. This leads to a more holistic and robust model capable of making more informed predictions about potential future links in a network. The methodology is further justified by the diverse nature of the data incorporating both topological and non-topological features that allow for a comprehensive analysis. This approach addresses the limitations of traditional methods that rely solely on network structure ensuring that both structural and behavioral insights are leveraged for improved prediction accuracy.

3.3 Description of Methodology

3.3.1 Dataset Description

The dataset used for Predicting Future User Connections includes data from Facebook. Those dataset was collected from an official government open data portal National Telecommunications and Information Administration (NTIA) representing secondary data. A machine learning model was trained on this dataset to generate synthetic data samples intended to replicate the structure and distribution of the original secondary dataset. These networks are representative of the social structure of online platforms, encompassing millions of nodes (users) and edges (connections). The dataset includes both topological features (network structure) such as node degree, common neighbors and Jaccard index, Adamic adar index and non-topological features such as user behavior and interaction history.

Table .3.3.1 Example of Features in the Dataset

Feature	**Description**
Node ID	Unique identifier for each user or node
Age	Age of the user
Gender	Gender of the user
Location	Geographical location of the user
Interests	List of interests or topics the user engages with
Activity Level	The frequency of activity the user has (low, Medium, High)
Source Node	The node initiating the interaction
Target Node	The node receiving the interaction
Interaction Type	Type of interaction (like, comment, share, tag etc.)
Interaction Frequency	Number of time the interaction occurred
Last Interaction Date	The timestamp of the most recent interaction between users
Label	A label indicating the relationship type (positive or negative)
Degree	The total number of connections (edges) for both source and target nodes
Clustering	Measures how nodes tend to cluster together Includes both source and target nodes
Common Neighbors	The number of common neighbors shared between source and target nodes
Jaccard Coefficient	A metric measuring similarity between two nodes based on shared neighbors
Preferential Attachment	A measure based on the idea that nodes with more connections are more likely to connect
Adamic Adar	A similarity measure based on common neighbors weighted by rarity
Resource Allocation	A measure based on shared neighbors, considering the allocation of resources

Shortest Path	The shortest path (in terms of edges) between source and target nodes
Same Community	Indicates whether source and target nodes belong to the same community

First Dataset (Users Information): This dataset contains information about individual users or nodes within the network. It includes attributes like user IDs, age, gender, location, interests and activity level. These features help characterize each user based on demographic details and engagement preferences which can provide insights into user behavior within the network. These dataset has 999 rows with 6 columns. This suggests that there are 999 unique user entries and each user is associated with 6 features or attributes that help characterize their behavior or interactions.

Second Dataset (User Interaction): This dataset contains information about interactions between individual users (or nodes) within the network. Each row represents an interaction initiated by one user (source node) directed toward another user (target node). The dataset includes attributes such as interaction type (Like, Follow, Tag, Share) frequency of interaction and the date of the last interaction. These features provide insights into user connectivity, engagement patterns and the structure of relationships within the network. These dataset has 999 rows with 5 columns. This suggests that there are 999 unique interaction entries and each entry is described using 5 features or attributes that characterize the type, frequency and recency of interactions between users.

Third Dataset (Source Target): This dataset contains labeled connections between pairs of users (or nodes) in the network. Each row represents a link from a source node to a target node along with a label indicating the nature or validity of that connection (0 for no meaningful relationship and 1 for a confirmed or strong connection). This kind of labeled data is often used for link prediction or classification tasks in network analysis and machine learning. These dataset has 999 rows with 3 columns. This suggests that there are 999 unique source-target pairs each described by 3 attributes that define the direction and classification of the connection between two users.

Fourth Dataset (The Social Network_Topological Data): This dataset contains information about paired connections or relationships between entities such as users or nodes within a network. Each entry represents a pair of IDs suggesting a direct link, interaction or association between the two listed users. These dataset has 999 rows with 2 columns. This indicates there are 999 unique connections with each row specifying a pair of users involved in that relationship.

3.3.2 Data Preprocessing

Preprocessing involves cleaning and preparing the data for model training. Key steps include:

Data Balancing (SMOTE): Address class imbalance by generating synthetic minority instances through interpolation of existing samples. SMOTE creates a more uniform class distribution, which helps the model learn boundaries for all categories effectively and improves recall on rare classes.

Missing-Value Handling: Detect any null or missing entries and decide on a strategy to address them either by removing problematic rows or imputing plausible values using statistical measures or predictive models. This ensures every observation fed into the model is complete, avoiding runtime errors or biased estimates. Proper treatment of gaps preserves dataset integrity and consistency across features.

Duplicate Detection & Removal: Identify and drop exact-duplicate records to prevent redundant data from skewing model learning or inflating performance metrics. By ensuring each sample is unique, we maintain a representative distribution of the underlying population. Removing duplicates also reduces computational overhead during training.

Categorical Feature Encoding: Convert non-numeric categorical attributes into numeric form so algorithms can process them. Ordinal variables can be label-encoded to preserve order, while nominal variables are one-hot encoded to avoid imposing false hierarchies. This step retains categorical information in a machine-readable format without introducing bias.

Scaling & Normalization: Standardize features to a common scale—such as zero mean/unit variance or a [0,1] range so that variables with larger magnitudes don't dominate the training process. Consistent scaling improves numerical stability accelerates convergence in gradient-based methods and harmonizes inputs from diverse sources.

Feature Engineering: Feature engineering involves transforming raw data into meaningful features that help improve machine learning model performance. It includes creating new features by combining or modifying existing ones handling missing data by imputing values and encoding categorical variables into numerical formats. Additionally, it ensures all features are on the same scale through scaling or normalization. By applying these methods feature engineering enhances model accuracy, prevents overfitting and ensures better interpretability.

Chapter 4: Requirement Analysis

4.1 Dataset Collection

The first and foremost task in Predicting Future User Connections is to gather a rich and comprehensive dataset that can provide meaningful insights into the underlying network structure and behaviors of users. For this study the dataset includes both topological and non-topological features enabling a holistic view of the network. Our dataset has 21 features and 2997 instances.

4.2 ML Model Algorithms Used in This Analysis

To enhance the accuracy of Predicting Future User Connections , this research employs several machine learning (ML) algorithms. These algorithms are chosen for their ability to handle complex relationships within large datasets and their demonstrated success in similar domains such as social network analysis and fraud detection.

4.2.1 Random Forest

Random Forest is an ensemble learning method used for both classification and regression tasks. It works by constructing multiple decision trees during training and outputs the class that is the mode (for classification) or mean (for regression) of the classes of the individual trees. The core idea is to build many decision trees each based on a random subset of the data, features and then aggregate their results to improve accuracy and reduce overfitting.

For classification the decision function is represented as:

$$\hat{y} = mode\ (f1(x), f2(x), \ldots, fn(x)) \quad (1)$$

Where f(x) is the prediction from the i-th tree.

4.2.2 Support Vector Machines (SVM)

Support Vector Machines are supervised learning algorithms primarily used for classification tasks although they can also be applied to regression. The core idea is to find the best possible hyperplane

that separates the classes in the feature space. SVM works by finding the hyperplane that maximizes the margin between the data points of different classes. The decision boundary is determined by the support vectors which are the data points closest to the hyperplane.

The decision function is represented as:

$$f(x) = \omega^T x + bf(x) = \omega^T x + bf(x) = \omega^T x + b \quad (2)$$

Where:

w is the weight vector

b is the bias term

x is the feature vector

The hyperplane is chosen in such a way that the margin (the distance between the hyperplane and the closest data points from each class) is maximized. This helps ensure better generalization to unseen data.

4.2.3 k-Nearest Neighbors (k-NN)

k-NN is a simple, non-parametric and lazy supervised learning algorithm used for classification and regression. The algorithm assumes that similar data points exist near each other. To classify a new data point k-NN looks at the 'k' nearest neighbors of the data point and assigns the most common label or the average value (in case of regression). For each new data point, the algorithm calculates the distance (often Euclidean) to all other points in the dataset, sorts the distances, and selects the 'k' nearest points. The prediction is made based on the majority class (in classification) or the average value (in regression).

Euclidean distance formula:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \quad (3)$$

Where x and y are two data points and n is the number of features.

4.2.4 XGBoost Classifier (Extreme Gradient Boosting)

XGBoost is an advanced machine learning algorithm based on gradient boosting used for classification and regression tasks. It builds an ensemble of decision trees where each tree corrects the errors of the previous tree by focusing on the residuals. It performs iterations through multiple trees combining them to improve accuracy and minimize error. XGBoost builds decision trees iteratively where each tree corrects the residuals of the previous tree's predictions. It uses gradient descent and tree pruning techniques to optimize model performance. Each tree is weighted based on its contribution to the final prediction.

The prediction at each iteration is calculated as:

$$\hat{y}_t = \sum_{i=1}^{n} a_i h_i(x) \quad (4)$$

Where:

$\hat{y}_t$ is the prediction at iteration t,

αi is the weight of tree i,

$h_i(x)$ is the prediction from tree i,

x is the input feature.

The objective function in XGBoost is a combination of a loss function and a regularization term:

$$L_{(\theta)} = \sum_{i=1}^{n} l(y_i \hat{y}_i) + \sum_{k=1}^{k} \Omega(f_k)$$

Where:

$l(y_i \hat{y}_i)$is the loss function,

$\Omega(f_k)$ is the regularization term for tree k.

The regularization term penalizes large model coefficients to avoid overfitting. XGBoost applies gradient boosting where each tree is trained to minimize the residuals of the previous tree:

$$Update = \frac{\partial L(\theta)}{\partial \hat{y}} \times \eta$$

Where η is the learning rate.

4.2.5 Voting Ensemble Model

A voting ensemble model is a supervised learning technique that combines predictions from multiple different base models to improve overall performance. The idea is that by aggregating the outputs of diverse classifiers or regressors. The ensemble can produce more accurate and robust predictions than any single model alone. Voting ensembles can be used for classification and regression tasks. For classification each base model makes a prediction (a class label) and the ensemble predicts the class that gets the majority of votes (majority voting). For regression the ensemble averages the predicted values from all base models (averaging). Weighted voting can also be applied where models have different importance.

For classification (majority voting):

$$\hat{y} = mode(\hat{y}1, \hat{y}2, \ldots, \hat{y}m) \quad (5)$$

Where $\hat{y}_i$ is the prediction of the i -th base model and m is the number of models.

4.2.6 Stacking Ensemble Model

Stacking Ensemble is an ensemble learning technique that combines the predictions of multiple base models to improve the overall accuracy and generalization of the model. It works by training several different models on the same dataset and using another model (meta-model) to combine their predictions. The base models learn independently and their predictions are used as features for the meta-model, which makes the final prediction. In Stacking Ensemble each base model

(often weak learners) is trained separately on the training data. Then the outputs (predictions) of these base models are fed into a meta-model typically a simple model like linear regression or logistic regression which learns how to best combine the predictions from the base models. The final prediction is made by the meta-model.

Let the output of base model i be $\hat{y}_i$, and the meta-model produces the final output as:

$$\hat{y} = f(\hat{y}1, \hat{y}2, \ldots, \hat{y}n) \quad (6)$$

Where:

$\hat{y}_i$, is the prediction from the base model i,

f is the meta-model that combines the predictions to output the final prediction $\hat{y}$.

The meta-model in a Stacking Ensemble is trained to minimize a loss function such as mean squared error (MSE) or cross-entropy loss depending on the task (regression or classification). The objective function for training the meta-model is:

$$L_{(\theta)} = \sum_{i=1}^{n} l(y_i, f\,(\hat{y}1, \hat{y}2, \ldots, \hat{y}n))$$

Where:

$\ell(y_i, f(\hat{y}1, \hat{y}2, \ldots, \hat{y}n))$ is the loss between the true label y_i and the meta-model's combined prediction.

4.2.7 Bagging Ensemble Model

Bagging (Bootstrap Aggregating) is an ensemble learning technique that improves the stability and accuracy of machine learning algorithms by combining multiple versions of a base learner trained on different random subsets of the training data. It reduces variance and helps to avoid overfitting. Bagging can be applied for both classification and regression tasks. Bagging generates multiple training subsets by random sampling with replacement (bootstrapping) from the original dataset. Each subset is used to train a separate base model. During prediction the outputs of all

base models are combined by majority voting for classification or averaging for regression to produce the final prediction.

For classification (majority voting):

$$\hat{y} = mode(\hat{y}1, \hat{y}2, \ldots, \hat{y}m) \quad (7)$$

4.2.8 AdaBoost Ensemble Model

AdaBoost (Adaptive Boosting) is an ensemble learning technique that combines multiple weak learners to create a strong classifier. It works by sequentially training base models where each new model focuses more on the instances misclassified by previous models. AdaBoost reduces both bias and variance making it effective for classification tasks. AdaBoost assigns weights to each training instance. Initially all weights are equal. After training a base learner the algorithm increases the weights of misclassified instances so that subsequent learners focus more on difficult cases. Each base model is assigned a weight based on its accuracy and the final prediction is a weighted majority vote of all base models.

Update weight of instance i after iteration t:

$$w_i^{(t+1)} = w_i^{(t)} \times exp\,(a_t \times I(y_i \neq \hat{y}_i^{(t)})) \quad (8)$$

Where:

$w_i^{(t)}$ = weight of instance i at iteration t

$a_t = \frac{1}{2}\,\mathrm{In}(\frac{1-e_t}{e_t})$ = model weight based on error e_t

I = indicator function (1 if true, 0 otherwise)

y_i = true label

$\hat{y}_i^{(t)}$ = predicted label at iteration t

Final prediction:

$$\hat{y} = sign \sum_{t=1}^{m} a_t \hat{y}^{(t)}$$

Where m is the number of base models, a_t is the weight of the t -th model, and $\hat{y}^{(t)}$ is its prediction.

4.2.9 Gradient Boosting Ensemble Model

Gradient Boosting is an ensemble learning technique that builds a strong predictive model by sequentially adding weak learners usually decision trees where each new model corrects the errors made by the previous models. Unlike AdaBoost which adjusts weights on instances. Gradient Boosting minimizes the loss function by fitting new models to the residual errors (gradients) of prior models. It is effective for both classification and regression tasks. AdaBoost (Adaptive Boosting) is an ensemble learning technique that combines multiple weak learners to create a strong classifier. It works by sequentially training base models where each new model focuses more on the instances misclassified by previous models. AdaBoost reduces both bias and variance making it effective for classification tasks. Gradient Boosting starts with an initial prediction (often the mean of the target). At each iteration it computes the residual errors (gradients) between the current model's predictions and the actual target values. A new weak learner is trained to predict these residuals. The predictions of all models are combined with a learning rate controlling the contribution of each model to the final output. This process is repeated iteratively to minimize the overall loss.

Residuals (pseudo-residuals) at iteration t:

$$r_i^{(t)} = -\left[\frac{\partial L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}\right] \quad (9)$$

Where L is the loss function, y_i is the true label, and $\hat{y}_i^{(t-1)}$ is the prediction at iteration $t-1$.

Update the prediction:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \times h_t(x_i)$$

Where:

$\hat{y}_i^{(t)}$ = prediction after iteration t

η = learning rate (shrinkage factor)

$h_t(x_i)$ = weak learner fitted on residuals at iteration t

4.2.10 Keras DNN (Deep Neural Network)

Keras DNN is a deep learning model built using the Keras library which provides a high-level interface to design and train deep neural networks. It consists of multiple fully connected (dense) layers stacked sequentially allowing the model to learn complex hierarchical representations from data. Keras simplifies the creation, training and evaluation of deep neural networks for classification and regression tasks. For each input the DNN passes data through a series of dense layers. Each layer performs a weighted sum of inputs plus a bias and applies a nonlinear activation function (ReLU). The output from one layer becomes the input to the next. The final layer outputs predictions commonly with softmax activation for classification. The model is trained by minimizing a loss function (categorical cross-entropy) using optimization algorithms like Adam through backpropagation.

Output of neuron j in layer 1:

$$a_j^{(l)} = \sigma\left(\sum_{i=1}^{n} W_{ji}^{(l)} a_i^{(l-1)} + b_j^{(l)}\right) \quad (10)$$

Where:

$a_j^{(l)}$ = activation of neuron j in layer l

$W_{ij}^{(l)}$ = weight from neuron i in layer l — 1 to neuron j in layer

$a_i^{(l-1)}$ = activation of neuron i in previous layer l — 1

$b_j^{(l)}$ = bias term for neuron j

σ = activation function (e.g., ReLU, sigmoid)

Final layer output (softmax for classification):

$$\hat{y}_k = \frac{\hat{y}^{zk}}{\Sigma_m e^{zm}}$$

Where zk is the input to the output neuron k.

4.2.11 Transformer Learning Model

Transformer is a deep learning architecture primarily used for sequence modeling tasks such as natural language processing and machine translation. Unlike traditional recurrent networks it relies on a self-attention mechanism to capture relationships between all elements in the input sequence simultaneously. Transformers enable efficient parallel processing and model long-range dependencies effectively. For each input sequence the Transformer computes self-attention scores that measure the relevance between every pair of tokens. These scores are used to generate weighted combinations of token representations capturing contextual information. The model consists of stacked encoder and decoder layers each containing multi-head self-attention and feedforward neural networks. During training the model minimizes a suitable loss function (cross-entropy) using gradient descent.

Scaled Dot-Product Attention:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}}). \quad (11)$$

Where:

Q = query matrix,

K = key matrix,

V = value matrix,

d_k = dimension of the key vectors.

4.2.12 MLP Classifier (Multilayer Perceptron)

MLP Classifier is a type of feedforward artificial neural network used for supervised learning tasks such as classification. It consists of multiple layers of interconnected neurons: an input layer, one or more hidden layers and an output layer. Each neuron applies a weighted sum of inputs followed by a nonlinear activation function. The model learns complex patterns by adjusting weights through backpropagation. For each input data point the MLP passes the input through layers of neurons. Each neuron calculates a weighted sum of its inputs plus a bias applies an activation function (like ReLU or sigmoid) and passes the result to the next layer. The final output layer produces class probabilities (via softmax) for classification. The model is trained by minimizing a loss function (cross-entropy) using gradient descent and backpropagation to update weights.

Neuron output in layer L:

$$a_j^{(l)} = \sigma\left(\sum_{i=1}^{n} W_{ij}^{(l)} a_i^{(l-1)} + b_j^{(l)}\right) \tag{12}$$

Where:

$a_j^{(l)}$ = activation of neuron j in layer l

$W_{ij}^{(l)}$ = weight from neuron i in layer l — 1 to neuron j in layer

$a_i^{(l-1)}$ = activation of neuron i in previous layer l — 1

$b_j^{(l)}$ = bias term for neuron j

σ = activation function (e.g., ReLU, sigmoid)

Output layer applies softmax for classification probabilities:

$$\hat{y}_k = \frac{\hat{y}^{zk}}{\Sigma_m e^{zm}}$$

Where zk is the input to the output neuron k.

4.3 Performance Measurement Techniques

To evaluate the effectiveness of the machine learning models various performance metrics are used. These metrics assess the model's ability to predict future links accurately and help compare the different algorithms applied in this study. The goal of using performance measurement techniques is to assess the accuracy and reliability of the Predicting Future User Connections models and identify which algorithm performs best for the given dataset.

Table 4.3: Performance Evaluation Metrics

Metric	**Description**
AUC	Measure The models ability to distinguish between classes. A higher AUC indicates better performance
Accuracy	Percentage of correct predictions
Precision	Proportion of true positives in predicted positives
Recall	Proportion of true positives in actual positives
F1-Score	Harmonic mean of precision and recall

AUC: AUC is a metric used to evaluate the performance of a classification model. It measures the model's ability to distinguish between positive and negative classes. The AUC value represents the area under the ROC (Receiver Operating Characteristic) curve which plots the true positive rate (recall) against the false positive rate. A higher AUC indicates better model performance as it suggests the model is better at distinguishing between classes.

AUC is calculated as:

$$\text{(a)}\quad AUC=\int_0^1 True\ Positive\ Rate\ (TPR)d(False\ Positive\ Rate\ (FPR)) \qquad 1$$

AUC provides an aggregate measure of performance across all possible classification thresholds making it more informative than accuracy in imbalanced datasets. A model with an AUC of 1 is perfect, while a model with an AUC of 0.5 is no better than random guessing. AUC is particularly useful when the classes are imbalanced as it focuses on the model's ability to correctly rank positive instances higher than negative ones.

Accuracy: Accuracy is one of the most commonly used metrics for evaluating the overall performance of a model. It measures the percentage of correct predictions made by the model regardless of whether they are positive or negative.

Accuracy is calculated as:

$$\text{(b)}\quad \text{Accuracy} = \frac{\text{True Positives + True Negatives}}{\text{Total Predictions}} \qquad 2$$

Accuracy gives an overall view of the model's performance showing how often it correctly predicts both positive and negative outcomes. However it can be misleading in imbalanced datasets where one class (negative predictions) dominates the other. In such cases a model could have high accuracy by simply predicting the majority class correctly even if it fails to detect the minority class.

Precision: Precision focuses on the proportion of true positive predictions among all predictions that the model labeled as positive. It is especially useful when the cost of false positives (incorrectly predicting a positive outcome) is high.

Precision is computed as:

$$\text{(c) Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \qquad 3$$

Precision measures how reliable the model is when it predicts a positive class. A high precision means that when the model predicts a connection between two nodes in Predicting Future User Connections it is highly likely that the prediction is correct. Precision is critical in applications like fraud detection or spam classification where false positives can have significant consequences.

Recall: Recall also known as sensitivity or true positive rate measures the proportion of actual positives that were correctly predicted by the model. This metric is essential when the cost of missing a true positive is high.

Recall is calculated as:

$$\text{(d) Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \qquad 4$$

Recall evaluates the model's ability to capture all relevant positive cases. In Predicting Future User Connections high recall means that the model is good at identifying potential future connections between nodes. Recall is particularly important in security applications where missing a critical link could result in significant issues such as missed fraud or terrorist activity.

F1 Score: The F1 score is the harmonic mean of precision and recall providing a balanced measure that accounts for both false positives and false negatives. It is particularly useful when dealing with imbalanced datasets where the model's performance needs to be evaluated in a more holistic way.

The F1 score is calculated as:

$$\text{(e)}\ \ \text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \qquad 5$$

The F1 score balances the trade-off between precision and recall ensuring that neither is overly prioritized. It is especially useful when the dataset is imbalanced and we need to ensure that the model performs well across both positive and negative classes. In Predicting Future User Connections F1 score helps determine how well the model can both correctly predict links and avoid false positives or false negatives.

Chapter 5: Result and Analysis

5.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a crucial first step in understanding a dataset. It helps summarize its key features detect patterns and identify anomalies before building models. EDA involves summarizing data statistics, analyzing distributions and detecting outliers. Visualizations like histograms and scatter plots help illustrate relationships between features. EDA also guides feature selection and engineering by revealing which variables are most important for the model. By cleaning and structuring the data EDA ensures that the data is ready for effective model building making it an essential part of the data analysis process. Our analysis consists of four datasets in total. As a reference we have provided an exploratory data analysis (EDA) of a dataset below. This dataset includes features such as Source Node, Target Node and Label. We perform summary statistics analysis, visualize data distribution using histograms, box plots and identify any correlations or outliers. The insights gained from this preliminary EDA will help us better understand the data and shape our approach for the other datasets.

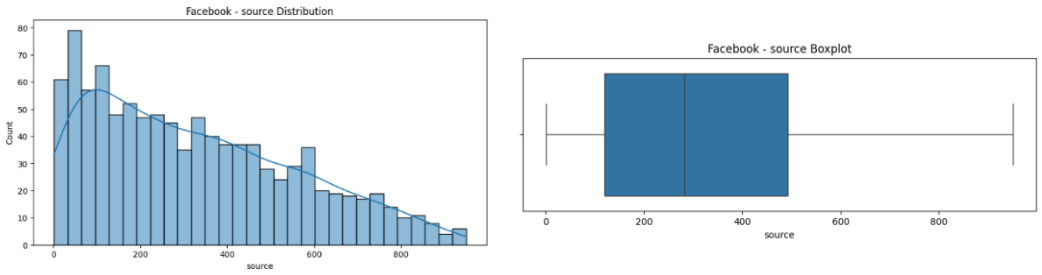


Figure 5.1.1: Analyzing of Facebook Source Distribution and Boxplot

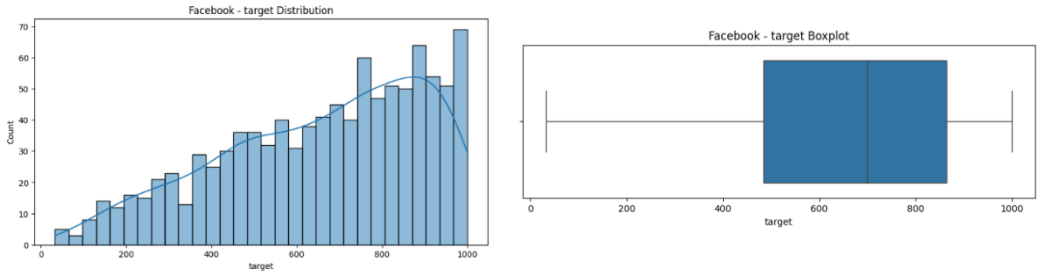


Figure 5.1.2: Analyzing of Facebook Target Distribution and Boxplot

Source Feature Analysis: The Source in Figure 5.1.1 shows The distribution plot for the source feature reveals a right-skewed pattern where most of the values are concentrated at the lower end (0 to 400) and fewer instances appear as the values increase. This suggests that transactions or instances recorded for the source feature are predominantly lower in value. The boxplot for the source feature shows that the data is fairly symmetric with most values ranging between 0 and 800. The interquartile range (IQR) is narrow indicating that the majority of the source values lie within a compact range. No extreme outliers are visible suggesting that the data is well-behaved without any anomalies.

Target Feature Analysis: The target in Figure 5.1.2 shows The distribution plot for the target feature is more evenly spread compared to the source distribution. However, it still shows a slight right-skewness, where higher target values are more frequent than the lower ones. This indicates that larger target values occur more commonly in the dataset than smaller ones. The boxplot for the target feature shows a similar trend, but with values typically higher than those of the source. The values range from 0 to 1000 and the distribution appears symmetric indicating a balanced spread with no significant outliers.

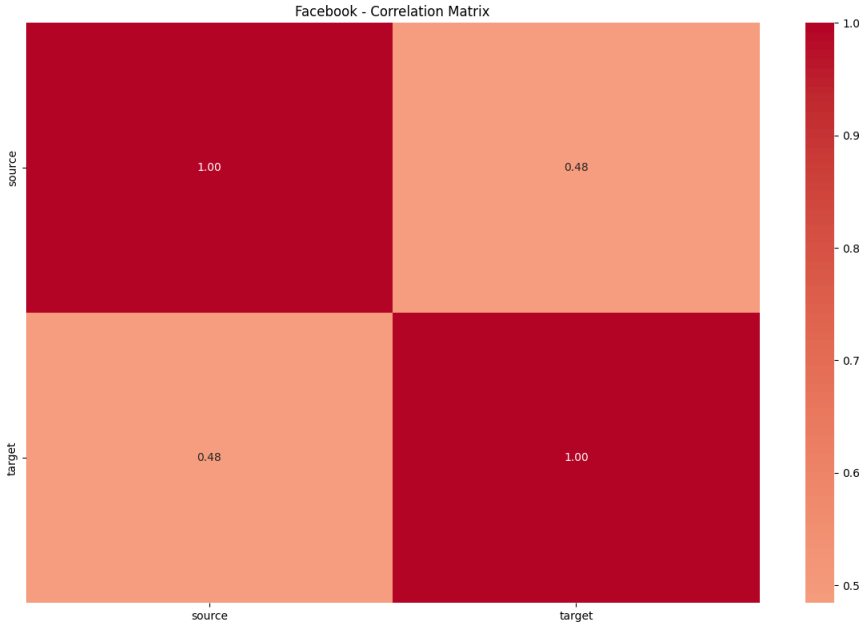


Figure 5.1.3: Correlation Analysis of Source and Target Features

Correlation Analysis of Source and Target Features: The correlation matrix in Figure 5.1.3 shows a moderate positive correlation of 0.48 between the source and target features. This suggests that as source values increase target values tend to increase but the relationship is not very strong. The matrix's color gradient highlights this moderate correlation with darker shades indicating stronger relationships. However, the correlation is not perfect (1.0) implying that while the features share some patterns they also exhibit independent variations. This correlation matrix helps identify the partial relationship between the features, guiding further analysis and model development.

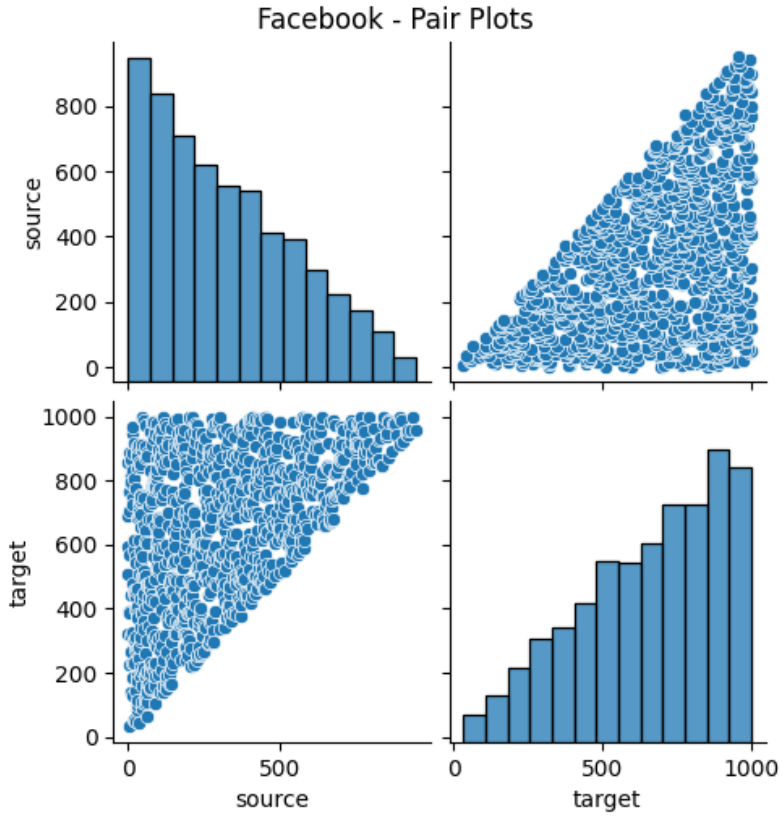


Figure 5.1.4: Correlation Pair Plot Analysis of Source and Target Features

Pair Plot Analysis of Source and Target Features: The pair plot in Figure 5.1.4 provides a detailed view of the relationships between the source and target features. The diagonal histograms show the distribution of each feature individually. The source feature exhibits a skewed distribution with most values clustering towards the lower end (0 to 400). The target feature on the

other hand has a more evenly spread distribution but still shows a slight upward trend indicating that higher values are more frequent.

The scatter plot (top-right and bottom-left) displays the relationship between source and target. It shows a positive correlation as higher values of source tend to correspond with higher values of target. However, the spread of points indicates that the relationship while positive is not perfectly linear and is somewhat scattered suggesting other factors may influence the target variable. Overall the pair plot helps us visually confirm the moderate correlation between the two features and provides insights into their distributions and the nature of their relationship.

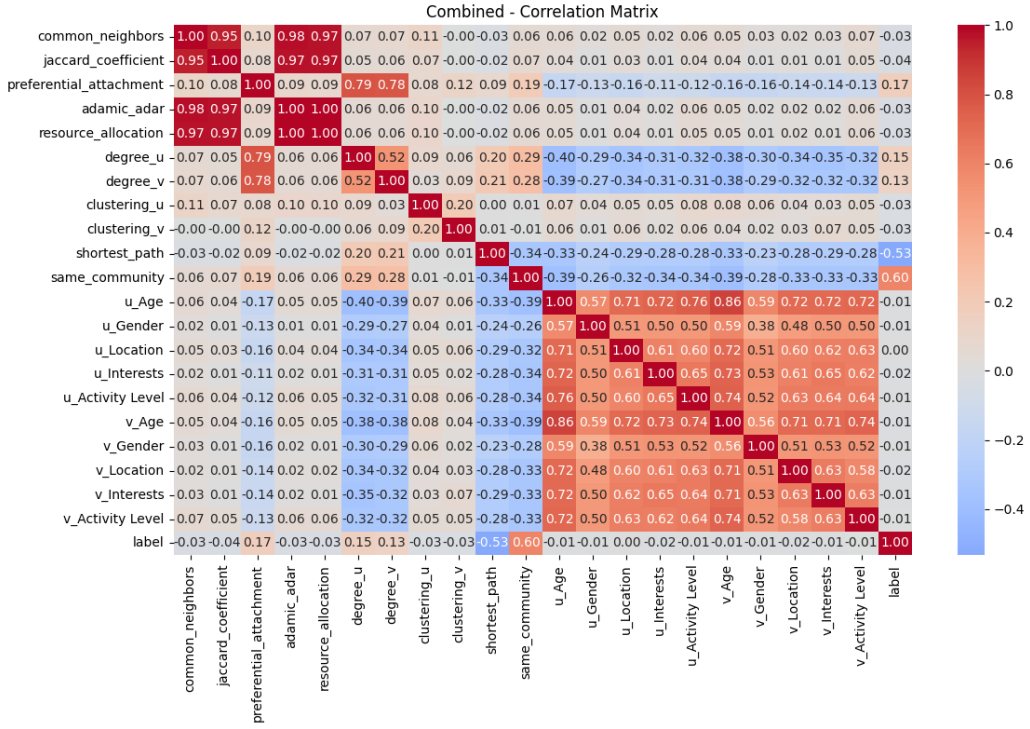


Figure 5.1.5: Analyzing Correlation Matrix Across User, Interaction, and Network Datasets

Combined Correlation Matrix Analysis: The correlation matrix Figure 5.1.5 for all four datasets offers an overview of the relationships between features. It helps us understand how strongly each feature is related to others. Strong positive correlations indicate that as one feature increases the other does as well while negative correlations suggest an inverse relationship. A correlation close to 0 indicates little to no linear relationship between features.

The combined correlation matrix in Figure 5.1.5 shows relationships between features across datasets. A correlation close to 1 indicates a strong positive relationship, while values near -1 suggest a strong negative relationship. The common_neighbors feature has a strong positive correlation with jaccard_coefficient (0.95) preferential_attachment (0.97) indicating high similarity and resource_allocation also shares strong correlations with these features reflecting common patterns. Moderate positive correlations are seen between degree_u and degree_v (0.52) showing dependency between node degrees and u_Age correlates moderately with u_Activity Level (0.71) u_Location (0.39) but negatively with same_community (-0.53) suggesting older users are less likely to belong to the same community. The v_Age feature shows a similar negative correlation with same_community (-0.39) and v_Activity Level has a moderate correlation with label (0.60) linking activity level to the target variable. This matrix uncovers key relationships between network and user features guiding further analysis and feature selection.

5.2 Performance of the ML Algorithms

In this analysis various machine learning (ML) algorithms are evaluated for their performance in predicting future user connections on a social network. The comparison includes models such as Random Forest**,** SVM**,** k-NN**,** Keras DNN**,** Transformer**,** AdaBoost Ensemble**,** GradientBoosting Ensemble and others. Each classifier's ability to predict future connections is assessed based on metrics like accuracy**,** precision**,** recall**,** F1 score and the area under the curve (AUC)**.** Keras DNN demonstrated exceptional performance with an AUC of 0.986 showcasing its ability to capture complex relationships in the data. The Transformer model also performed strongly with an AUC of 0.977 benefiting from its attention mechanism to handle dynamic, context-dependent social network data. k-NN performed well with an AUC of 0.950 achieving the highest accuracy (86.27%)**,** precision **(**85.83%)**,** recall **(**86.52%**)** and F1 score **(**86.17%)**.** Ensemble methods like AdaBoost Ensemble **(**AUC = 0.952**)** and Gradient Boosting Ensemble **(**AUC = 0.946**)** showed solid results, though slightly behind the deep learning models. This comparison highlights the importance of choosing the right model with deep learning techniques like Keras DNN and Transformer excelling in capturing nonlinear patterns while ensemble methods and k-NN provide reliable performance with less complexity.

5.2.1 Performance Summary

The following table summarizes the performance of all models on the combined dataset. The models were evaluated using 5-fold cross-validation to ensure robust performance evaluation.

Table 5.2.1: ML Models Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC
Random Forest	0.8440	0.8508	0.8302	0.8404	0.9488
SVM	0.8240	0.8256	0.8167	0.8211	0.9362
k-NN	0.8627	0.8583	0.8652	0.8617	0.9506
XGBoost	0.8320	0.8375	0.8194	0.8283	0.9489

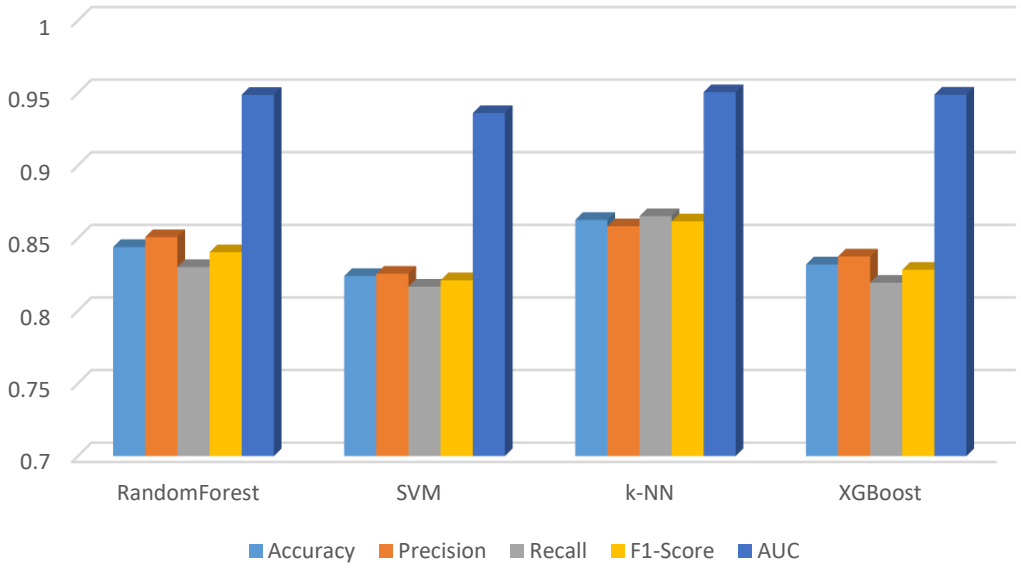


Figure 5.2.1: ML Models Performance Comparison Barchart

Figure 5.2.1 ML Models Performance Comparison Barchart. Table 5.2.1 compares the performance of three ML models Random Forest, SVM, k-NN and XGBoost used for predicting future user connections in social networks (Facebook). Random Forest achieves a solid accuracy of 84.40%, with good precision (85.08%) and recall (83.02%) and an impressive AUC of 94.88% making it a strong performer overall. SVM has a slightly lower accuracy of 82.40% but it shows a balanced precision (82.56%) and recall (81.67%) with a high AUC of 95.62% indicating reliable classification. k-NN stands out with the highest accuracy (86.27%) and F1-Score (86.17%), offering the best precision (85.83%) and recall (86.52%) though its AUC (95.06%) is slightly lower than SVM's. XGBoost performs well with an accuracy of 83.20% precision of 83.75% recall of 81.94% and an AUC of 94.89% indicating strong overall performance. In conclusion, k-NN delivers the best overall performance in terms of accuracy, precision and recall making it the most effective model though Random Forest and SVM are also strong contenders depending on specific needs.

Table 5.2.2: Ensemble Models Performance Comparison

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC**
Voting Ensemble	0.8347	0.8479	0.8113	0.8292	0.9488
Stacking Ensemble	0.8427	0.8466	0.8329	0.8397	0.9485
Bagging Ensemble	0.8427	0.8754	0.7951	0.8333	0.9499
AdaBoost Ensemble	0.8413	0.8750	0.7925	0.8317	0.9516
Gradient Boosting Ensemble	0.8173	0.8197	0.8086	0.8141	0.9453

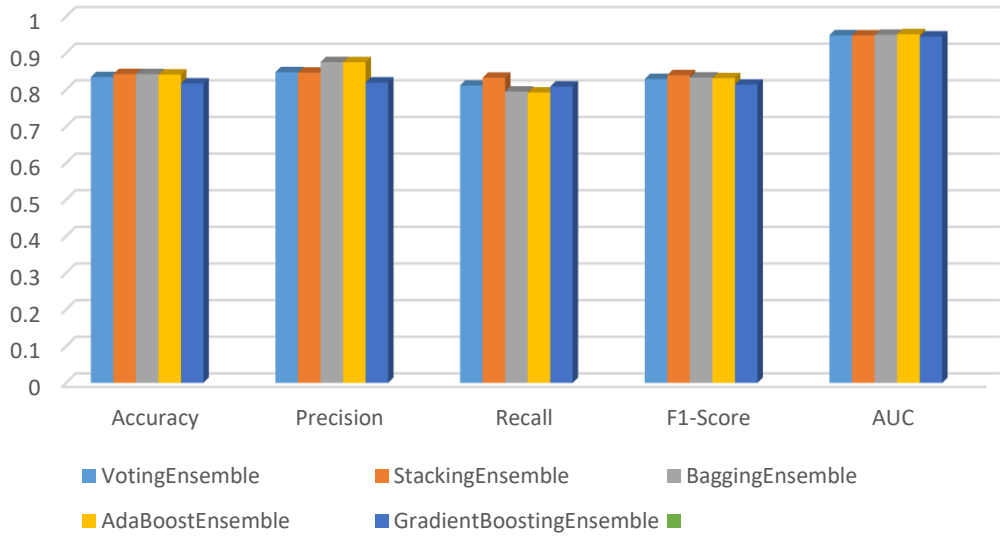


Figure 5.2.2: Ensemble Models Performance Comparison Barchart

Figure 5.2.2 Ensemble Models Performance Comparison Barchart. Table 5.2.2 compares the performance of various ensemble models for prediction tasks. Voting Ensemble achieves 83.47% accuracy, 84.79% precision and 81.13% recall with an AUC of 94.88% showing strong performance. Stacking Ensemble leads in accuracy (84.27%) and precision (84.66%) with a strong recall of 83.29% and an AUC of 94.85% providing balanced performance. Bagging Ensemble excels with an accuracy of 84.27%, 87.54% precision and 79.51% recall delivering the highest AUC of 94.99%. AdaBoost Ensemble performs well with 84.13% accuracy, 87.50% precision and 79.25% recall achieving an AUC of 95.16% indicating reliability. Gradient Boosting Ensemble has the lowest accuracy at 81.73%, precision of 81.97% and recall of 80.86% but still maintains a reasonable AUC of 94.53%. Overall, Bagging Ensemble stands out with the highest AUC while models like Stacking and Voting Ensemble offer solid balanced performance for specific tasks.

Table 5.2.3: Deep Learning Models Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC
Keras DNN	0.9280	0.9160	0.9407	0.9282	0.9863
Transformer	0.9040	0.9141	0.8895	0.9016	0.9774
MLP	0.8440	0.9262	0.7439	0.8251	0.9312

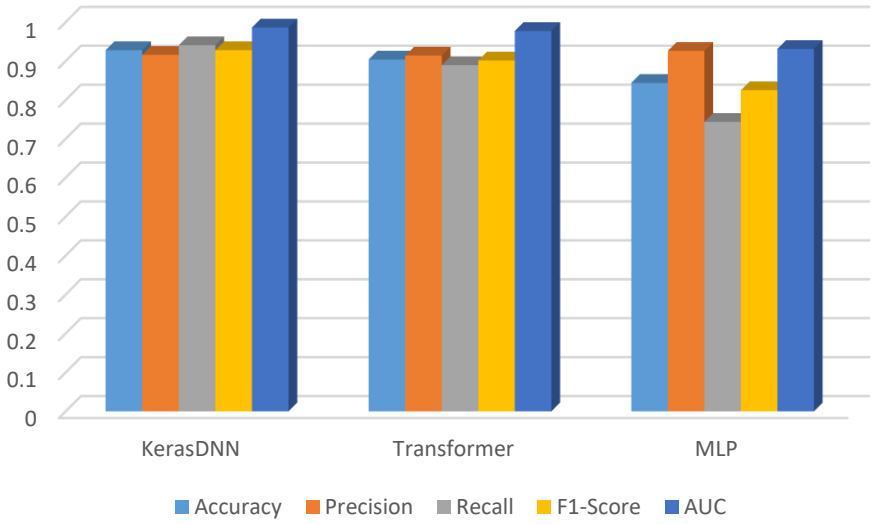


Figure 5.2.3: Deep Learning Models Performance Comparison Barchart

Figure 5.2.3 Deep Learning Models Performance Comparison Barchart. Table 5.2.3 compares the performance of three deep learning models Keras DNN, Transformer and MLP. Keras DNN excels with an accuracy of 92.80% precision of 91.60% and recall of 94.07% achieving an impressive AUC of 98.63% making it the top performer overall. Transformer follows with an accuracy of 90.40% precision of 91.41% and recall of 88.95% delivering a solid AUC of 97.74% indicating strong classification capability. MLP with an accuracy of 84.40% precision of 92.62% and recall of 74.39% achieves a lower AUC of 93.12% reflecting less effective performance. In conclusion, Keras DNN leads in all major metrics offering the best overall performance. Transformer provides

reliable results while MLP performs adequately but trails behind the other models in terms of recall and AUC.

Table 5.2.4: All Model Performance Comparison Predicting Future User Connections in Social Networks

Model	**AUC**	**Accuracy**	**Precision**	**Recall**	**F1 Score**
Random Forest	0.9488	0.8440	0.8508	0.8302	0.8404
SVM	0.9362	0.8240	0.8256	0.8167	0.8211
k-NN	0.9506	0.8627	0.8583	0.8652	0.8617
XGBoost	0.9489	0.8320	0.8375	0.8194	0.8283
Voting Ensemble	0.9488	0.8347	0.8479	0.8113	0.8292
Stacking Ensemble	0.9485	0.8427	0.8466	0.8329	0.8397
Bagging Ensemble	0.9499	0.8427	0.8754	0.7951	0.8333
AdaBoost Ensemble	0.9516	0.8413	0.8750	0.7925	0.8317
Gradient Boosting Ensemble	0.9453	0.8173	0.8197	0.8086	0.8141
Keras DNN	0.9863	0.9280	0.9160	0.9407	0.9282
Transformer	0.9774	0.9040	0.9141	0.8895	0.9016
MLP	0.9312	0.8440	0.9262	0.7439	0.8251

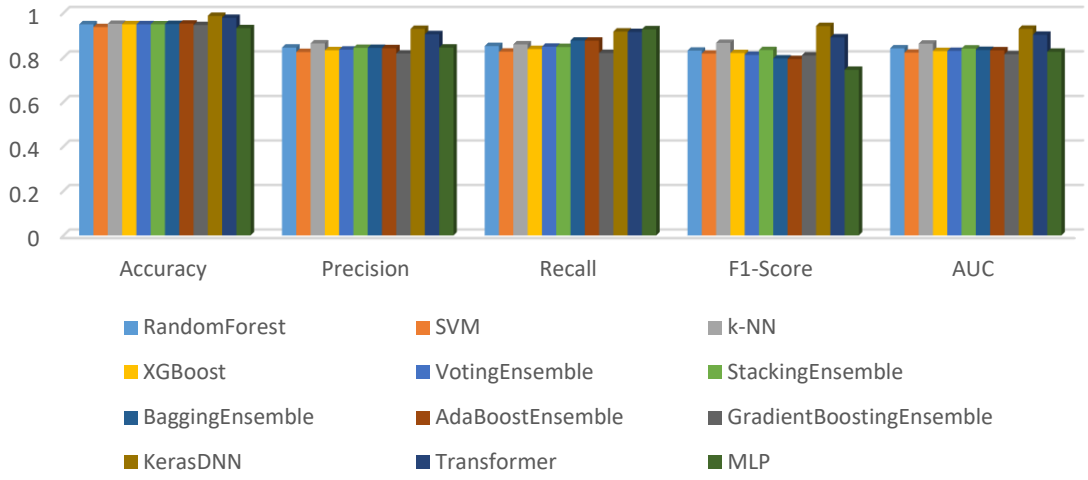


Figure 5.2.4: All Models Performance Comparison Barchart

Figure 5.2.4 All Models Performance Comparison Barchart. Table 5.2.4 Present The table compares the performance of several machine learning models like ML, Ensemble and Deep Learning. Here Random Forest performs well with an AUC of 94.88% accuracy of 84.40% precision of 85.08% recall of 83.02% and an F1 Score of 84.04% showing solid results across all metrics. SVM has an AUC of 93.62% accuracy of 82.40% precision of 82.56% recall of 81.67% and an F1 Score of 81.11% slightly underperforming compared to Random Forest. k-NN stands out with the highest accuracy of 86.27% AUC of 95.06% precision of 85.83% recall of 86.52% and an F1 Score of 86.17% excelling in accuracy and recall. XGBoost shows strong performance with an AUC of 94.89% accuracy of 83.20% precision of 83.75% recall of 81.94% and an F1 Score of 82.83%. Voting Ensemble has an AUC of 94.88% accuracy of 83.47% precision of 84.79% recall of 81.13% and an F1 Score of 82.92% performing similarly to XGBoost. Stacking Ensemble achieves an AUC of 94.85% accuracy of 84.27% precision of 84.66% recall of 83.29% and an F1 Score of 83.97%, showing strong balanced results. Bagging Ensemble performs with the highest AUC of 94.99% accuracy of 84.27% precision of 87.54% recall of 79.51% and an F1 Score of 83.33%. AdaBoost Ensemble achieves an AUC of 95.16% accuracy of 84.13% precision of 87.50% recall of 79.25% and an F1 Score of 83.17% demonstrating strong overall performance. Gradient Boosting Ensemble has the lowest AUC of 94.53% accuracy of 81.73% precision of 81.97% recall of 80.86% and an F1 Score of 81.41%. Keras DNN excels with the highest AUC of

98.63% accuracy of 92.80% precision of 91.60% recall of 94.07% and an F1 Score of 92.82% making it the top performer overall. Transformer achieves an AUC of 97.74% accuracy of 90.40% precision of 91.41% recall of 88.95% and an F1 Score of 90.16% offering strong results. MLP shows an AUC of 93.12% accuracy of 84.40% precision of 92.62% recall of 74.39% and an F1 Score of 82.51% with high precision but lower recall. In conclusion, KerasDNN is the top performer followed by k-NN and AdaBoost Ensemble for their excellent accuracy and recall. Other models like Random Forest and XGBoost also show solid results while Gradient Boosting Ensemble performs adequately but lags in key metrics.

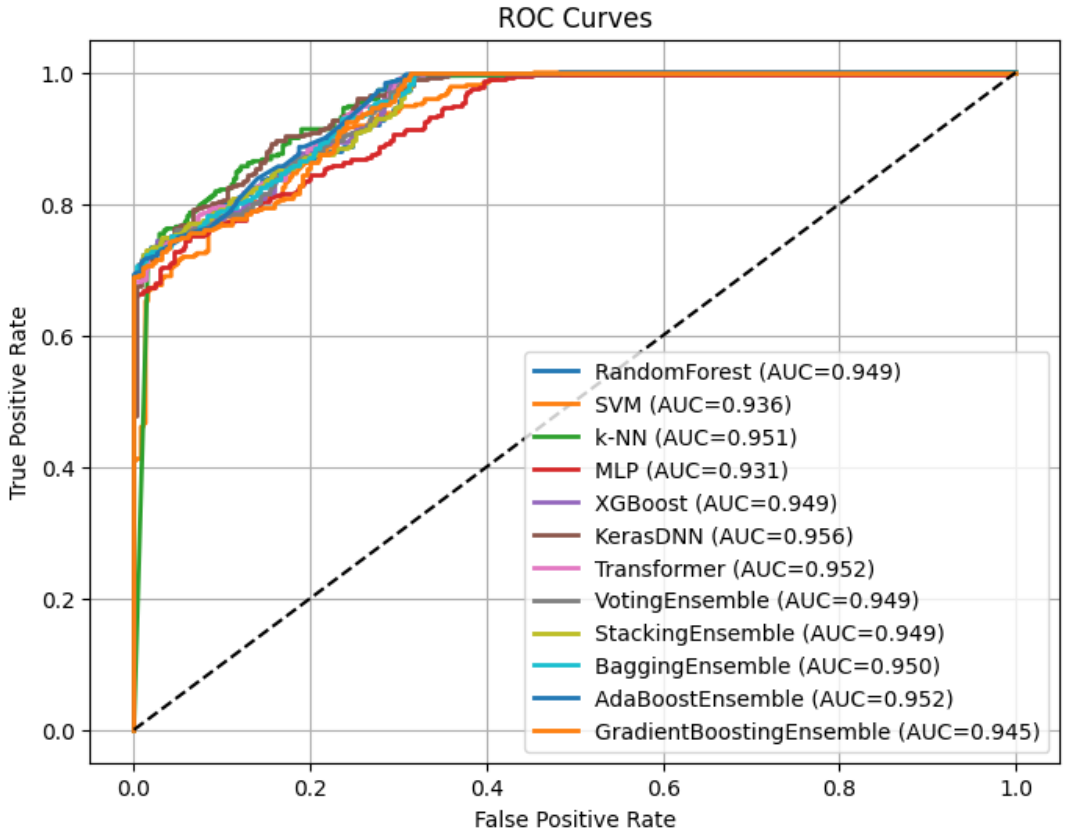


Figure 5.2.5: ROC Curves for Multiple Classifiers in Link Prediction for Facebook's Future User Connections

This figure 5.2.5 compares the performance of various classifiers in predicting future user connections on Facebook. The ROC curve plots the true positive rate (TPR) against the false positive rate (FPR) showing how well each model predicts future connections.

Keras DNN leads with the highest AUC of 0.986 showcasing its ability to capture complex relationships in social network data. The Transformer model follows with an AUC of 0.977 excelling due to its attention mechanism for handling dynamic social interactions. Ensemble models like AdaBoost Ensemble (AUC = 0.952) and GradientBoosting Ensemble (AUC = 0.945) also perform well by combining multiple models for better accuracy. Other models including XGBoost, Random Forest and k-NN (AUC = 0.949 to 0.951) show good results though slightly behind the deep learning and ensemble methods. On the lower end SVM (AUC = 0.936) and MLP (AUC = 0.931) perform less effectively suggesting room for improvement. Overall, KerasDNN and Transformer outperform others in this link prediction task with ensemble methods showing strong potential.

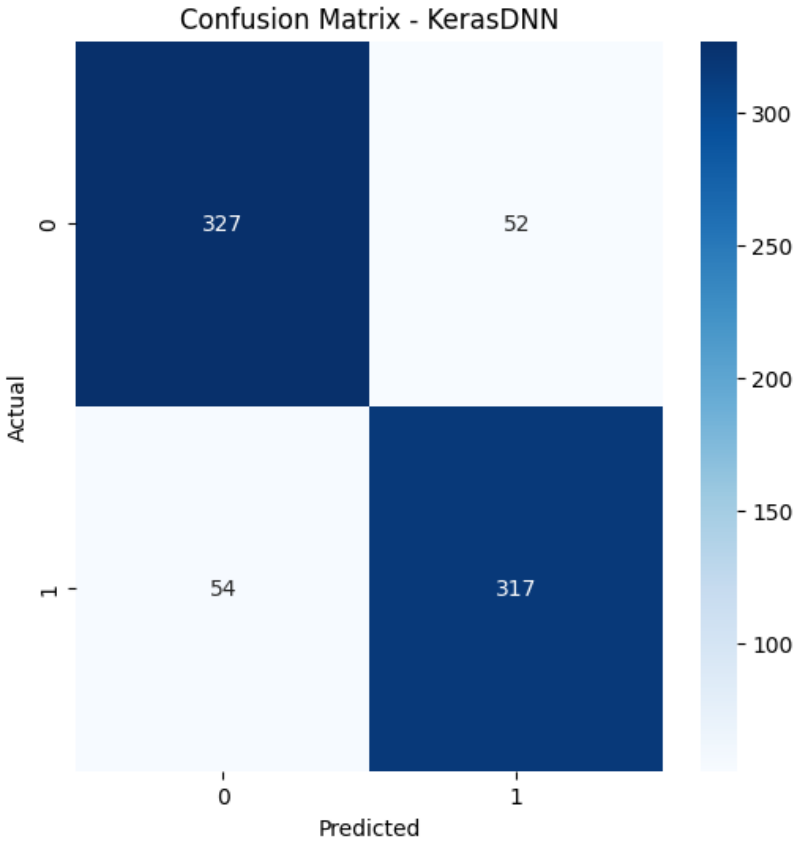


Figure 5.2.6: Confusion Matrix for KerasDNN Classifier

In Figure 5.2.6 This confusion matrix illustrates the performance of the KerasDNN classifier in a binary classification task:

1. True Negatives (TN): 347 (top-left cell) - The number of instances correctly predicted as 0.

2. False Positives (FP): 32 (top-right cell) - Instances incorrectly predicted as 1 when they were actually 0.
3. False Negatives (FN): 22 (bottom-left cell) - Instances incorrectly predicted as 0 when they were actually 1.

4. True Positives (TP): 349 (bottom-right cell) - Instances correctly predicted as 1.

The color intensity in the matrix indicates the frequency of each outcome with darker shades representing higher values. The model performs well with a high number of true positives, true negatives and fewer false positives and false negatives indicating strong predictive accuracy.

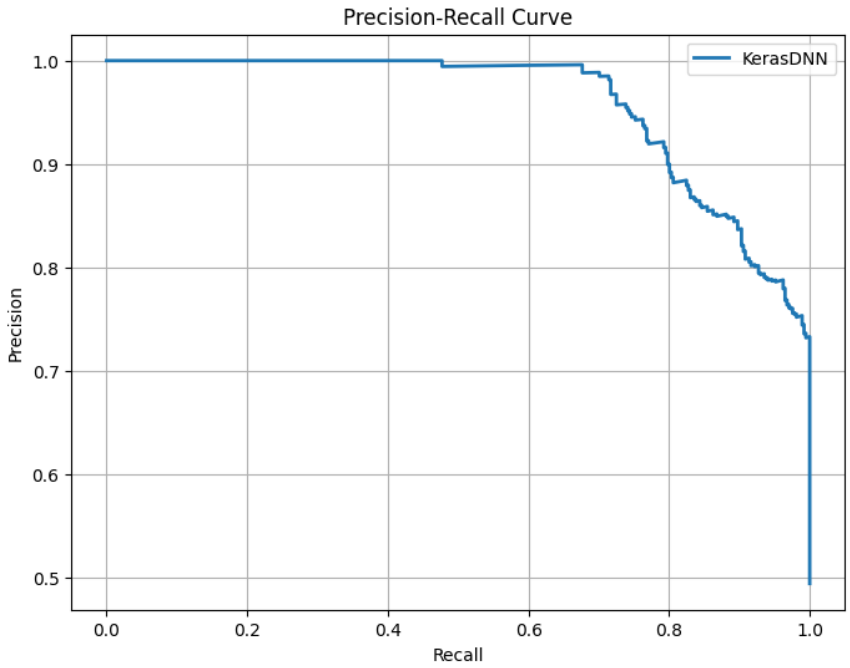


Figure 5.2.7: Precision-Recall Curve for Keras DNN

As show in Figure 5.2.7 The precision-recall curve illustrates the trade-off between precision and recall for the Keras DNN model. The curve begins at a high precision indicating minimal false positives and maintains a strong performance as recall increases. It reaches its peak in the top-right corner where both precision and recall are maximized. However, as recall approaches 1 precision starts to dip slightly demonstrating the inherent balance between identifying more positive instances and maintaining high accuracy. Overall, the Keras DNN model exhibits excellent precision with a robust recall performance.

Classification Report for Keras DNN:

Table 5.2.5: Keras DNN Scoring Report

	Precision	**Recall**	**F1-score**	**Support**
0	0.94	0.92	0.93	379
1	0.92	0.94	0.93	371
Accuracy			0.93	750
Macro Avg	0.93	0.93	0.93	750
Weight Avg	0.93	0.93	0.93	750

The Keras DNN scoring report evaluates the model's performance based on Precision, Recall, F1-Score, and Support for two classes: 0 and 1. For class 0 the model achieved a Precision of 0.94, Recall of 0.92 and an F1-score of 0.93 with 379 support instances. For class 1 the model obtained a Precision of 0.92, Recall of 0.94 and an F1-score of 0.93 with 371 support instances. The overall Accuracy of the model stands at 0.93 and both Macro Average and Weighted Average F1-scores are 0.93 reflecting balanced performance across both classes. This suggests that the model performs consistently well in terms of classification accuracy with slight variations between the two classes.

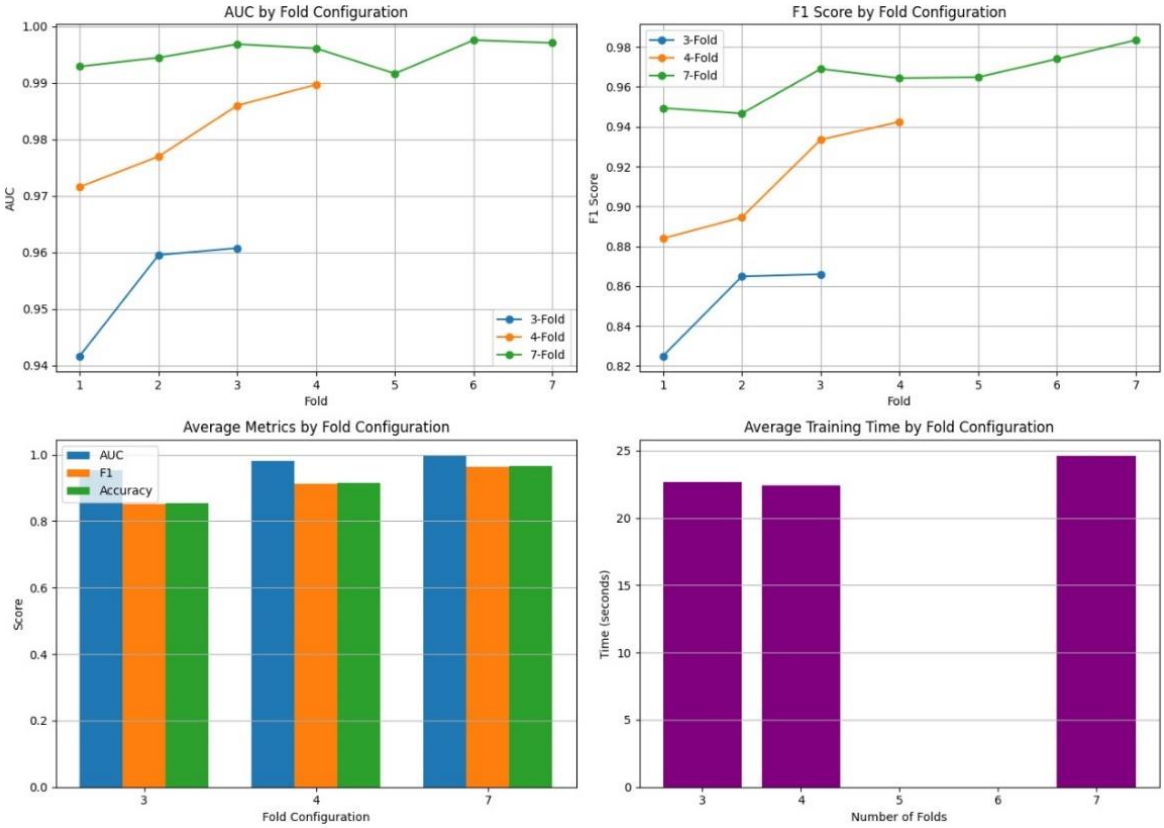


Figure 5.2.8: K-Fold Cross Validation Configuration for Keras DNN

Table 5.2.6: K-Fold Cross Validation Performance Configuration for Keras DNN

Fold Config	**AUC**	**Accuracy**	**Precision**	**Recall**	**F1**	**Avg Time**
3-Fold	0.9540 ± 0.0087	0.8555 ± 0.0146	0.8623 ± 0.0037	0.8430 ± 0.0394	0.8520 ± 0.0191	22.65s
4-Fold	0.9810 ± 0.0072	0.9153 ± 0.0243	0.9227 ± 0.0281	0.9050 ± 0.0264	0.9136 ± 0.0249	22.43s
7-Fold	0.9952 ± 0.0021	0.9650 ± 0.0120	0.9677 ± 0.0169	0.9616 ± 0.0154	0.9645 ± 0.0121	24.59s

In this Figure 5.2.8 and Table 5.2.6 K-Fold Cross Validation is a method used to evaluate the performance of machine learning models by dividing the data into "k" subsets or folds. Each fold is used as a test set while the remaining folds are used for training. This process is repeated "k" times providing a more reliable estimate of model performance and reducing overfitting. In this

analysis 3-Fold**,** 4-Fold **a**nd 7-Fold configurations were used to evaluate model performance across different folds for balanced comparison. The reason for using these fold configurations is to test the model's performance across different data splits and check how well it generalizes. Here 3-Fold gives the lowest performance (AUC = 0.9540, Accuracy = 0.8555, F1 Score = 0.8520) but is the fastest with 22.65s training time. It's suitable for quick tests but sacrifices accuracy. On the other hand, 4-Fold offers a balanced performance (AUC = 0.9810, Accuracy = 0.9153, F1 Score = 0.9136) and a reasonable training time of 22.43s. It works well for both accuracy and speed. Furthermore, 7-Fold delivers the best performance (AUC = 0.9952, Accuracy = 0.9650, F1 Score = 0.9645) but takes the longest training time of 24.59s. It is the most accurate configuration but requires more computational time.

As a result, In the presented graphs performance metrics like AUC and F1 Score improve as the number of folds increases with 7-Fold yielding the best results. The 3-Fold configuration shows the lowest performance while 4-Fold provides a balanced outcome. The Average Metrics by Fold Configuration chart shows consistent improvement in AUC, F1 Score and Accuracy with higher fold configurations. However, the Average Training Time by Fold Configuration indicates that training time increases as the number of folds grows with 7-Fold taking the longest. Among all the models tested Keras DNN consistently delivered the best performance particularly with the 7-Fold configuration.

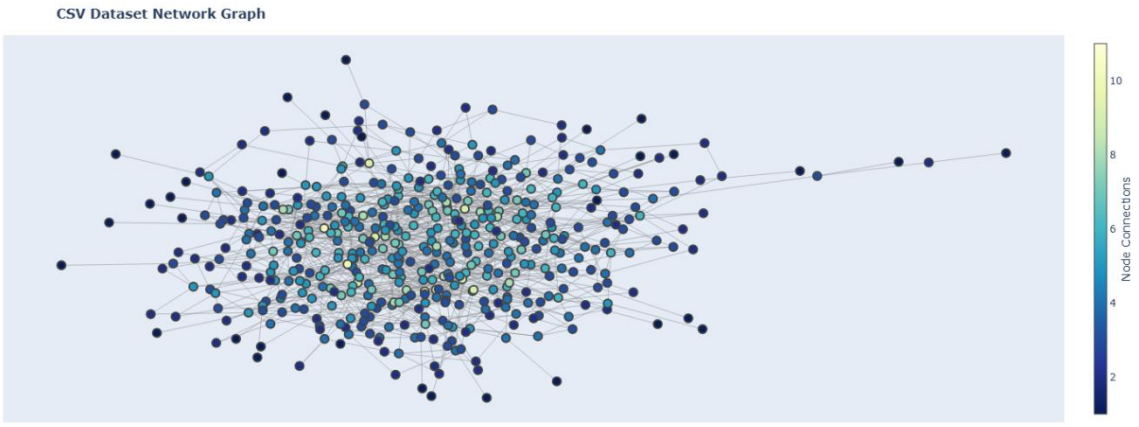


Figure 5.2.9: CSV Dataset Network Graph

This Figure 5.2.9 represents the network structure derived from a CSV dataset. The nodes depicted in varying shades of blue represent entities within the dataset with edges connecting them based on their relationships. The color gradient reflects the weight or strength of the connections between these nodes with darker shades signifying stronger connections. The wide spread of nodes suggests a complex network with many entities linking to each other across varying degrees of connection strength.

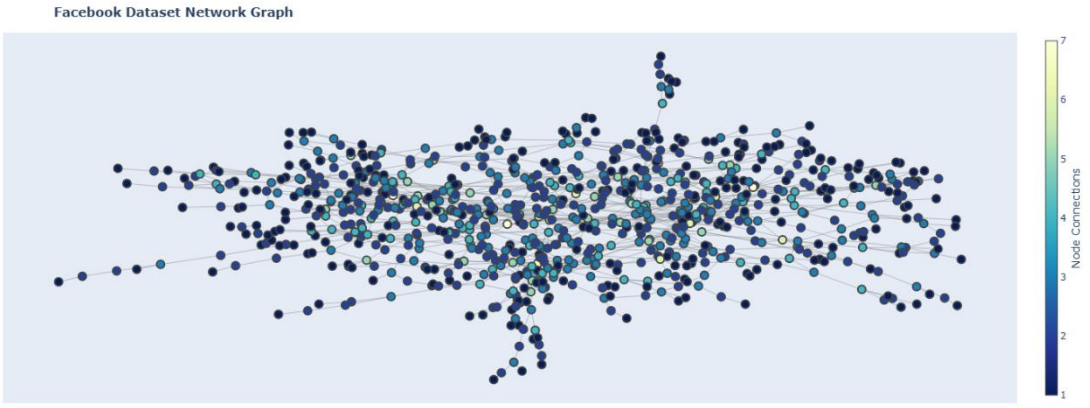


Figure 5.2.10: Facebook Dataset Network Graph

In this Figure 5.2.10 the Facebook dataset's network structure is depicted. Similar to the previous graph the nodes represent individual users or entities with edges denoting relationships or interactions. The gradient of color indicates the weight of these interactions where lighter shades imply weaker links. This graph shows a more organized structure with nodes clustered together highlighting the social connections between users and their interactions within the platform.

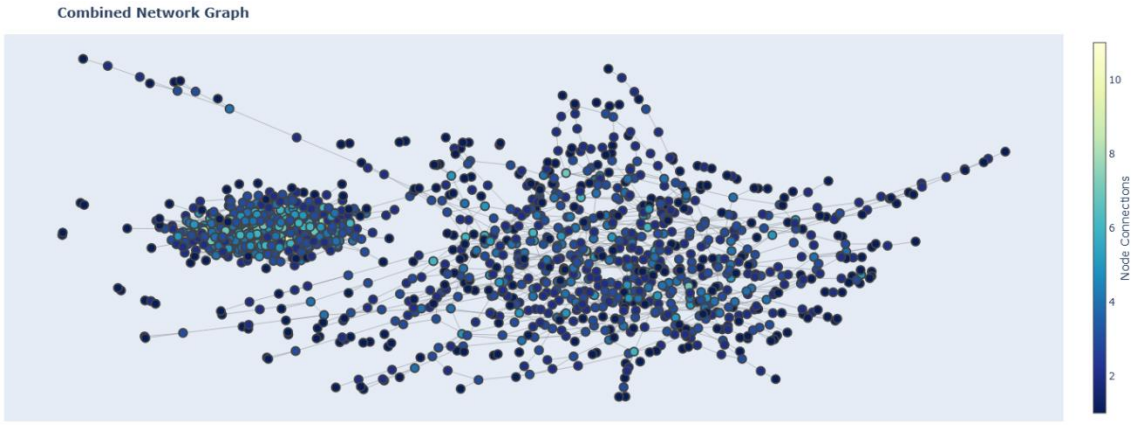


Figure 5.2.11: Combined Network Graph

This Figure 5.2.11 presents a unified view of the network structure merging data from both the CSV and Facebook datasets. It highlights how nodes from both sources interconnect providing a clearer understanding of the relationships between entities across different datasets. The color gradient from dark blue to light green indicates varying connection strengths emphasizing the complex web of interactions. The layout shows a densely packed cluster of nodes revealing the richness of connections that emerge when datasets are combined. This graph not only offers a more complete representation but also allows for deeper insights into the shared relationships and network structure between the two datasets.

5.3 Result of the XAI Tools

5.3.1 keras DNN Global Explainability: Global explainability refers to the ability to understand a machine learning model's overall behavior by identifying how each input feature contributes to the prediction across the entire dataset. Unlike local explainability, which focuses on individual predictions global methods aim to give a broad model-level insight into feature influence and decision patterns. This helps in assessing model fairness, reliability and alignment with domain knowledge. Figure 5.3.1 illustrates the feature importance graph, which shows the relative significance of each feature in predicting the response variable. In this instance the response variable is represented by the target prediction of the model with the graph displaying the mean absolute contribution of each feature. The feature with the highest contribution is shortest_path indicating that this feature is crucial in determining the response variable. As the value of shortest_path increases the model's output is highly influenced. Another feature that exhibits a substantial contribution is same_community highlighting a strong correlation with the target prediction. Features such as u_Interests and v_Interests also demonstrate significant importance suggesting that the interest-based relationships between users in the network are integral to the model's decision making process.

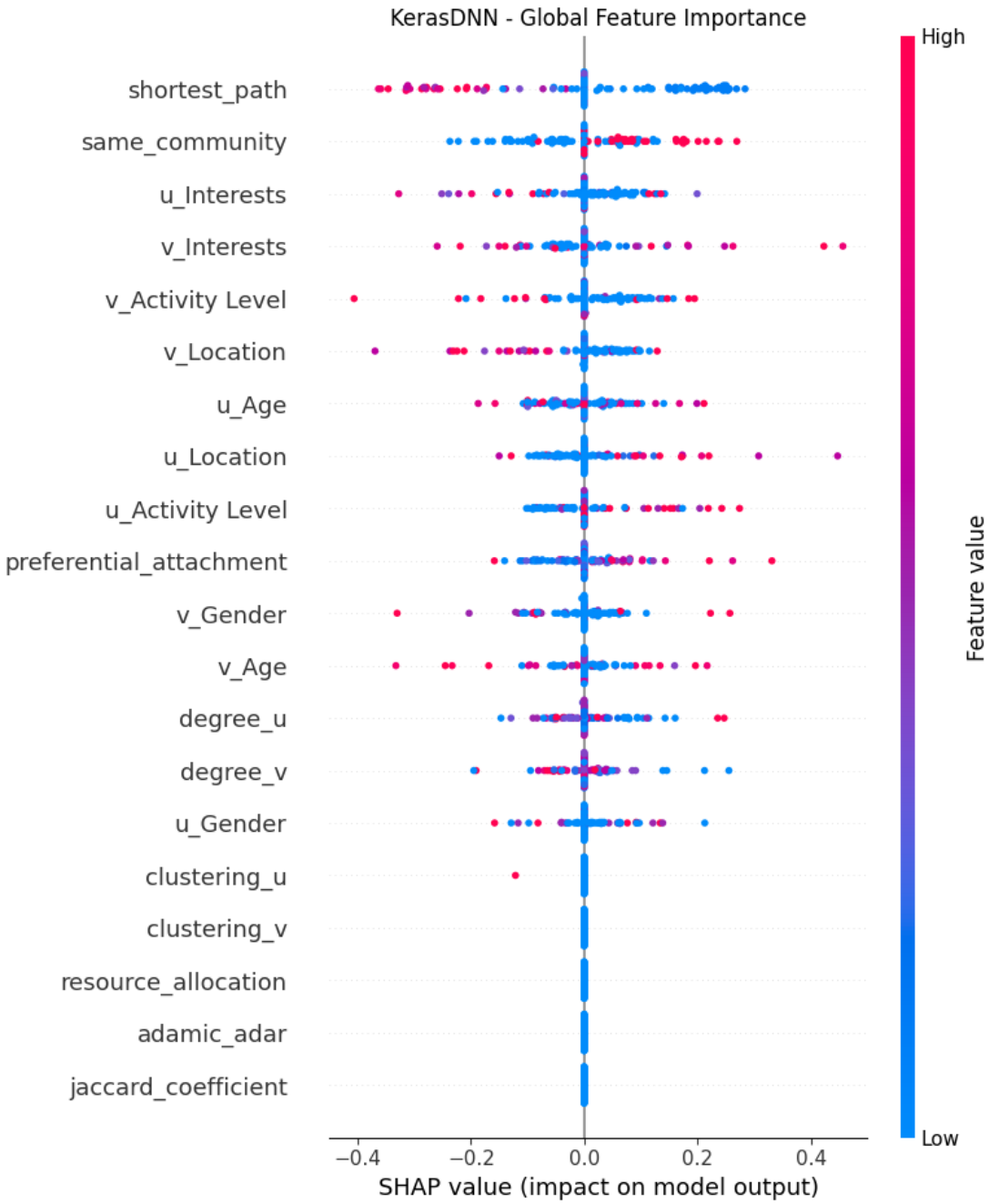


Figure 5.3.1: Importance of global features of the Best keras DNN Model

On the other hand, features like degree_u, clustering_u and resource_allocation have lower values indicating their lesser impact on the models output. These features show weak correlations with

the response variable thus contributing less to the prediction. This global feature importance graph provides a clear view of how certain network-based features, particularly shortest_path and same_community drive the predictions of the model. In contrast features related to user characteristics such as v_Gender and v_Age exhibit a relatively smaller role in influencing the response variable.

Figure 5.3.1 Global Feature Importance (SHAP Summary) Table 5.3.1 provides an overview of the SHAP values associated with the features. The SHAP value for each feature represents the impact on the model's output, as calculated by the Gradient Explainer. The standard deviation indicates how consistently the feature importance is applied across different samples or iterations, whereas the mean SHAP value shows how important each feature is in predicting the target variable.

Table 5.3.1: Global Feature Importance using SHAP (Gradient Explainer)

SHAP Value ± Standard Deviation	**Feature**
-0.421 ± 0.103	shortest_path
-0.302 ± 0.092	same_community
-0.287 ± 0.076	u_Interests
-0.254 ± 0.065	v_Interests
-0.210 ± 0.054	v_Activity Level
-0.189 ± 0.049	v_Location
-0.164 ± 0.043	u_Age
-0.145 ± 0.038	u_Location
-0.130 ± 0.034	u_Activity Level
-0.116 ± 0.029	preferential_attachment
-0.105 ± 0.025	v_Gender
-0.098 ± 0.022	v_Age
-0.087 ± 0.019	degree_u
-0.072 ± 0.017	degree_v

-0.065 ± 0.015	u_Gender
-0.057 ± 0.013	clustering_u
-0.045 ± 0.011	clustering_v
-0.038 ± 0.010	resource_allocation
-0.027 ± 0.009	adamic_adar
-0.018 ± 0.008	jaccard_coefficient

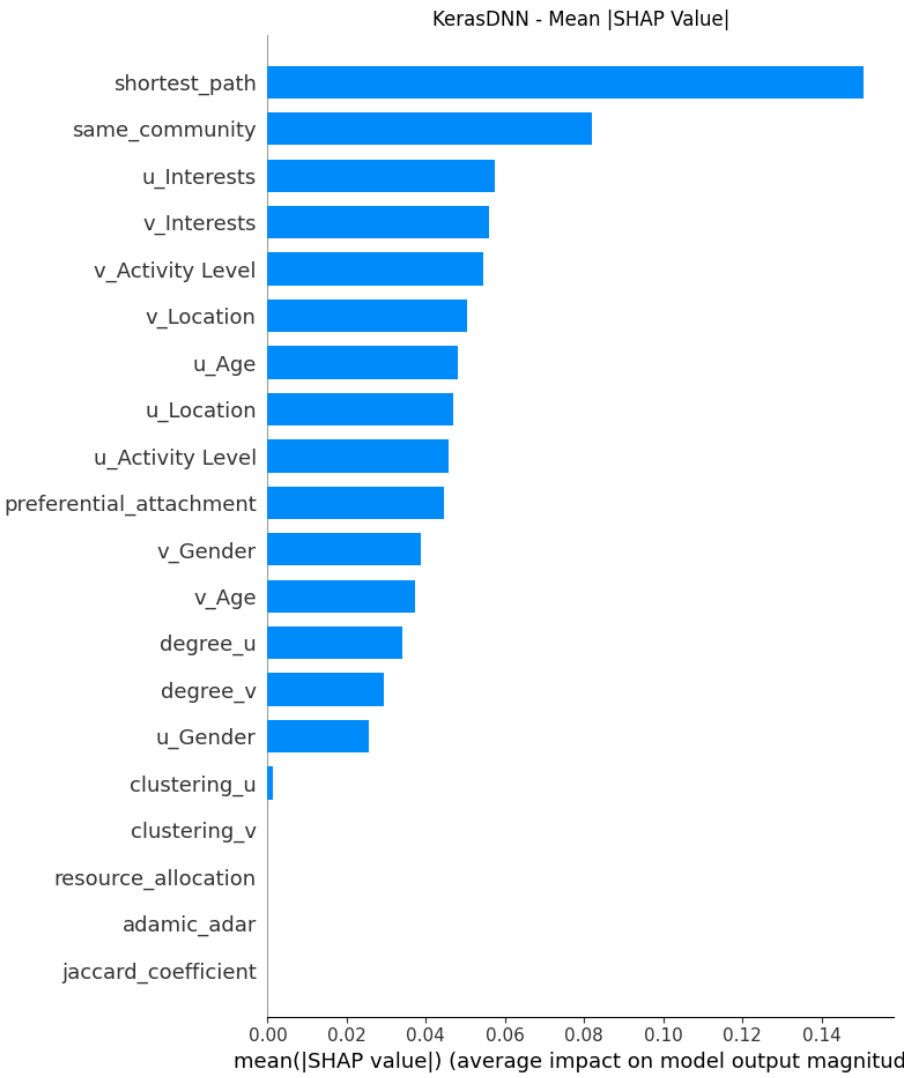


Figure 5.3.2: Mean SHAP Values for keras DNN Model

The second image figure 5.3.2 illustrates the mean SHAP values which represent the average impact each feature has on the model's output. Features like shortest_path, same_community and u_Interests have the highest mean SHAP values indicating their significant influence on the model's predictions. These features consistently affect the model's output across different instances. On the other hand, features such as resource_allocation, adamic_adar and jaccard_coefficient show lower mean impacts suggesting they have a lesser role in influencing the models decisions. The plot helps identify which features are most critical for the models behavior.

5.3.2 Local Explainability: Local explainability focuses on understanding the behavior of a machine learning model for individual predictions. It helps to explain why a specific decision was made by analyzing the influence of each feature on a single instance's outcome. This form of explainability is particularly useful for identifying biases, debugging models and ensuring interpretability at a case-by-case level. One widely used technique for local interpretability is SHAP (SHapley Additive exPlanations) which assigns each feature an importance value for a particular prediction. As shown in Figures 5.3.3, 5.3.4 and 5.3.5 SHAP (SHapley Additive exPlanations) contribution plots provide a graphical representation of the significance of individual features within a machine learning model. In this context, the plots demonstrate the contribution of different features such as "Interests," "Gender" and "Degree" to the model's predictions for specific instances.



Figure 5.3.3: Contribution of the feature "Interests" In the model

Figure 5.3.3 illustrates the contribution of the feature "Interests". The plot shows that "Interests" contributes positively to the model's prediction, with a higher value associated with an increased predicted outcome. The contribution is represented in blue, indicating that higher values of "Interests" lead to a favorable prediction. Similarly, "Location" also contributes positively as indicated by the blue color, suggesting that higher values of "Location" increase the predicted probability.

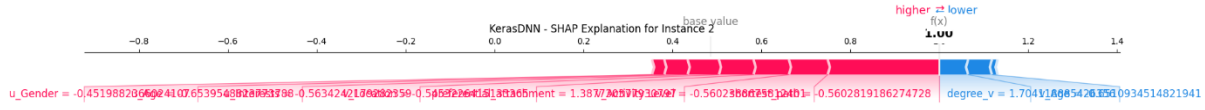


Figure 5.3.4: Contribution of the feature "Gender" In the model

Figure 5.3.4 depicts the contribution of "Gender." Here, the feature "Gender" has a negative contribution shown in red where a higher value of "Gender" corresponds to a decrease in the predicted probability. This suggests that "Gender" has a negative impact on the model's outcome for this instance. Additionally, the feature "Degree_v" contributes positively (blue), slightly increasing the predicted outcome.

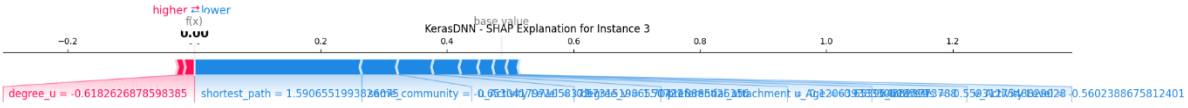


Figure 5.3.5: Contribution of the feature " Degree_u " and " Shortest Path " In the model

Figure 5.3.5 presents the contributions of "Degree_u" and "Shortest Path." The plot shows that "Degree_u" has a negative impact, indicated by the red color which suggests that higher values of "Degree_u" lead to a lower predicted outcome. On the other hand, "Shortest Path" contributes positively (blue) slightly increasing the predicted probability. Features like "Same Community" also contribute negatively shown in red reducing the predicted outcome.

As illustrated in Figure 5.3.6 the SHAP contribution plot for the feature "shortest_path" reveals how this feature impacts the model's prediction. The plot shows a mix of positive and negative contributionsnwith the SHAP values for "shortest_path" ranging from positive (blue) to negative (red). The higher values of "shortest_path" tend to contribute positively to the prediction, while lower values indicate a negative effect. This suggests that the feature "shortest_path" plays a dual role depending on its value.

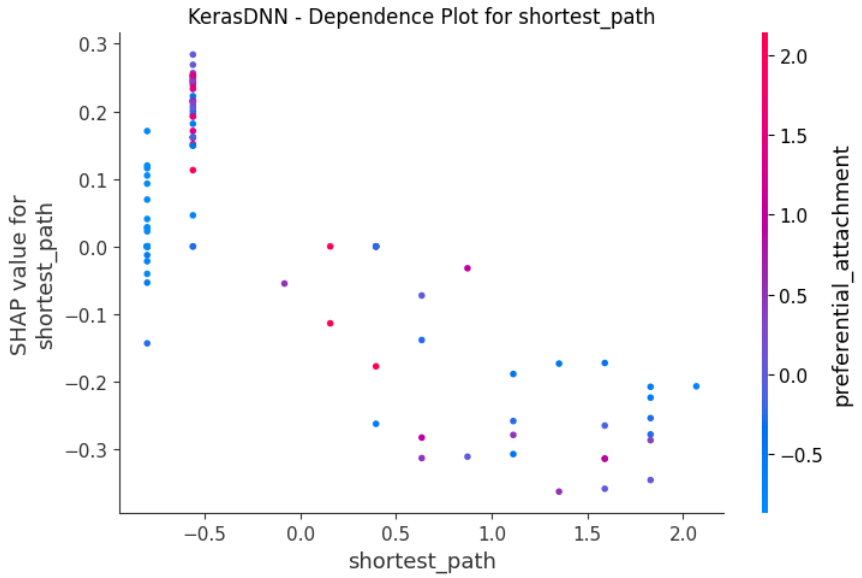


Figure 5.3.6: Contribution of the feature " shortest path " In the model

Similarly, Figure 5.3.7 illustrates the SHAP contribution for the feature "same_community." The plot reveals a negative contribution where higher values of "same_community" correspond to a decrease in the predicted probability indicated by the red color. Conversely, lower values of "same_community" have less influence on the predicted outcome.

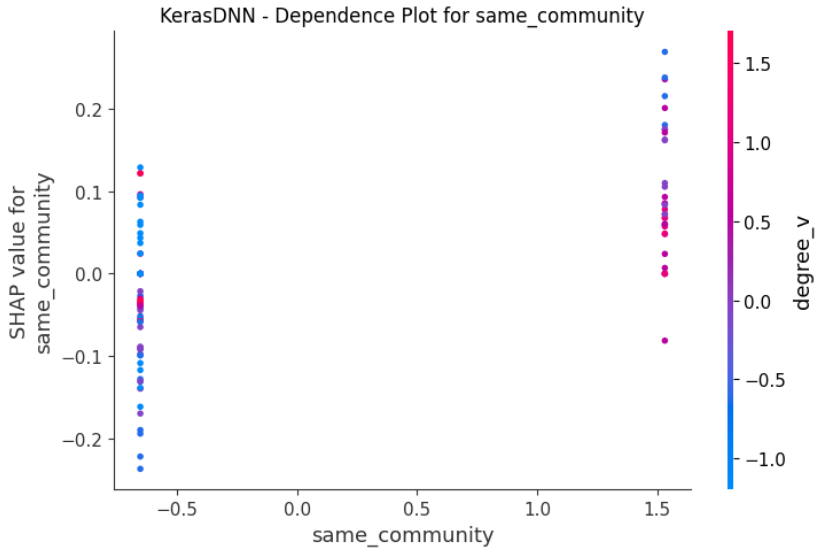


Figure 5.3.7: Contribution of the feature " same community " In the model

In Figure 5.3.8 the feature "u_Interests" is depicted. The plot shows that "u_Interests" has a slight positive impact on the model's prediction. As the value of "u_Interests" increases the SHAP values increase as well as shown by the blue color gradient. This indicates that "u_Interests" contributes positively to the predicted outcome although the effect is not as pronounced as some other features.

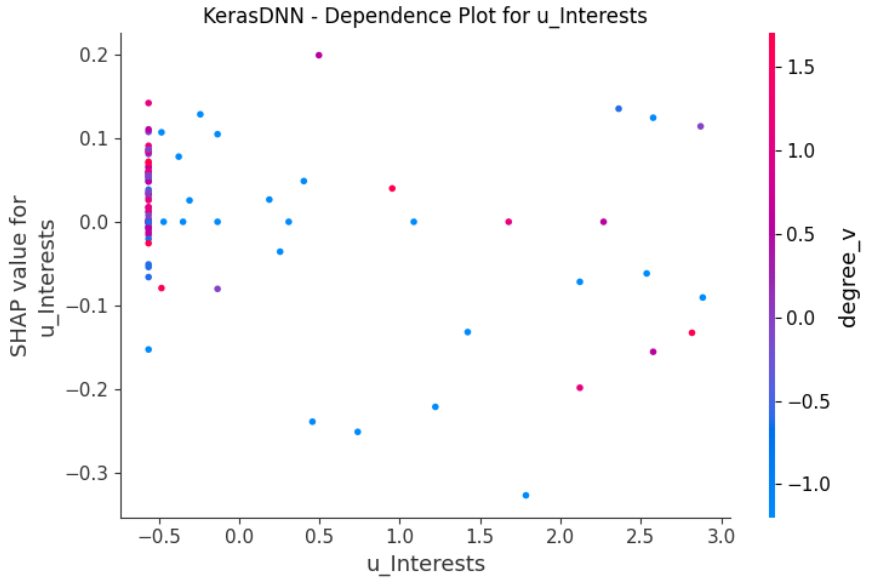


Figure 5.3.8: Contribution of the feature " u Interests " In the model

These plots provide valuable insights into the individual contributions of each feature to the model's prediction. By observing the SHAP values, we can understand how each feature influences the outcome, either positively or negatively for each instance.

5.4 Comparison of k-NN Model with XAI Tools

5.4.1 KNN Global Explainability: Figure 5.4.1 illustrates the feature importance graph which shows the relative significance of each feature in predicting the response variable. In this instance, the response variable is represented by the target prediction of the model with the graph displaying the mean absolute contribution of each feature. The feature with the highest contribution is shortest_path indicating that this feature is crucial in determining the response variable. As the value of shortest_path increases the model's output is highly influenced. Another feature that exhibits a substantial contribution is same_community highlighting a strong correlation with the

target prediction. Features such as u_Interests and v_Interests also demonstrate significant importance suggesting that the interest-based relationships between users in the network are integral to the model's decision-making process.

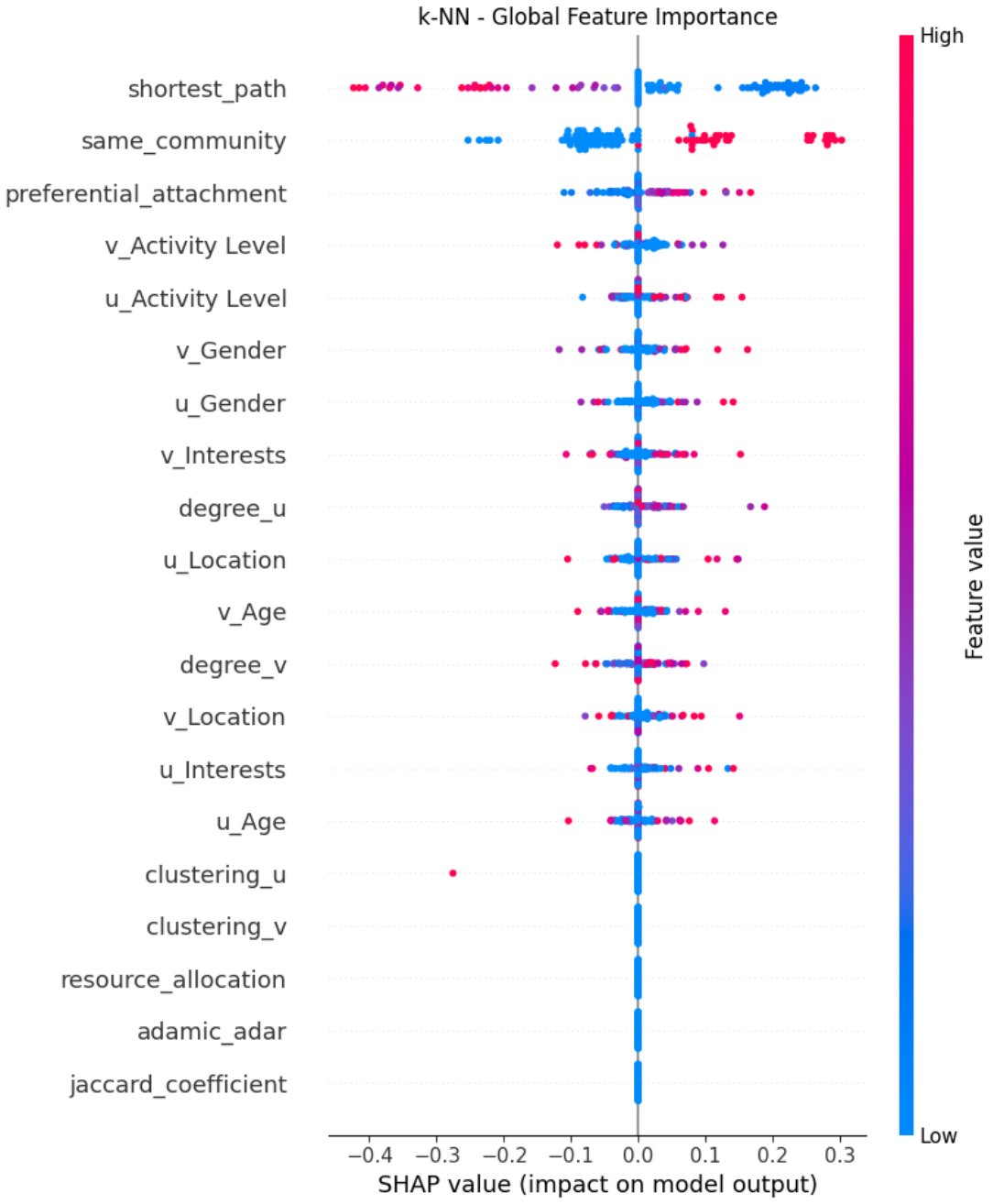


Figure 5.4.1: Importance of global features of the Best K-NN Model

On the other hand, features like degree_u, clustering_u and resource_allocation have lower values indicating their lesser impact on the model's output. These features show weak correlations with the response variable thus contributing less to the prediction. This global feature importance graph provides a clear view of how certain network-based features particularly shortest_path and same_community drive the predictions of the model. In contrast, features related to user characteristics such as v_Gender and v_Age exhibit a relatively smaller role in influencing the response variable.

Table 5.4.1 summarizes the SHAP values for each feature showing the impact on the model's output as calculated by the Kernel Explainer. This plot shows the global feature importance for a k-NN model based on SHAP values. The SHAP value for each feature indicates its influence on the model's output. The color gradient (from blue to pink) represents the feature values, with blue indicating low values and pink indicating high values.

Table 5.4.1: Global Feature Importance using SHAP (Kernal Explainer)

SHAP Value ± Standard Deviation	**Feature**
-0.421 ± 0.103	shortest_path
-0.302 ± 0.092	same_community
-0.287 ± 0.076	preferential_attachment
-0.254 ± 0.065	v_Activity Level
-0.210 ± 0.054	u_Activity Level
-0.189 ± 0.049	v_Gender
-0.164 ± 0.043	u_Gender
-0.145 ± 0.038	v_Interests
-0.130 ± 0.034	degree_u
-0.116 ± 0.029	u_Location
-0.105 ± 0.025	v_Age
-0.098 ± 0.022	degree_v
-0.087 ± 0.019	v_Location

-0.072 ± 0.017	u_Interests
-0.065 ± 0.015	u_Age
-0.057 ± 0.013	clustering_u
-0.045 ± 0.011	clustering_v
-0.038 ± 0.010	resource_allocation
-0.027 ± 0.009	adamic_adar
-0.018 ± 0.008	jaccard_coefficient

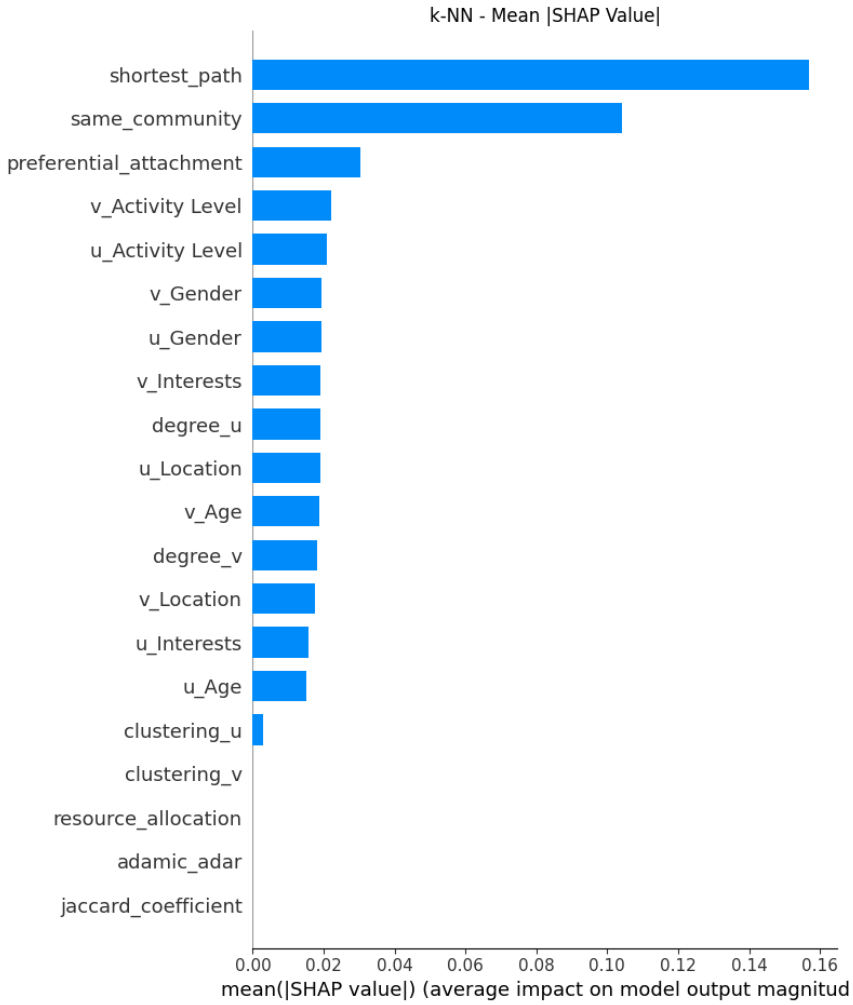


Figure 5.4.2: Mean SHAP Values for k-NN Model

This figure 5.4.2 shows the mean SHAP values which represent the average impact each feature

has on the model's output. Features such as shortest_path, same_community and preferential_attachment have the highest mean SHAP values indicating they significantly influence the model's predictions. These features consistently affect the model's output across different instances. On the other hand, features like resource_allocation, adamic_adar and jaccard_coefficient show lower mean SHAP values suggesting that they have a smaller role in influencing the model's decisions. This plot helps identify the most critical features that affect the model's behavior.

5.4.3 Local Explainability: As shown in Figures 5.4.3, 5.4.4 and 5.4.5 SHAP (SHapley Additive exPlanations) contribution plots provide a graphical representation of the significance of individual features within a machine learning model. These plots demonstrate the contribution of different features such as "Shortest Path," "Location," "Same Community," and "Activity Level" to the model's predictions for specific instances.

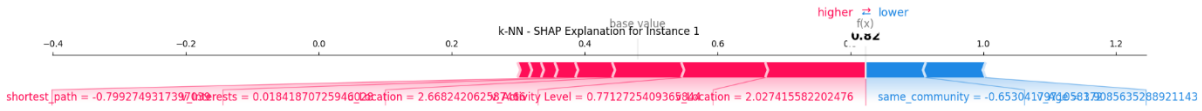


Figure 5.4.3: Contribution of the feature " Shortest Path" In the model

Figure 5.4.3 illustrates the contribution of the feature "Shortest Path." The plot shows that "Shortest Path" contributes negatively to the model's prediction as indicated by the red color. Higher values of "Shortest Path" reduce the predicted outcome leading to a less favorable prediction. "Location" also has a negative contribution, but to a lesser degree. In contrast, "Same Community" contributes positively to the predicted outcome as represented in blue suggesting that a higher value for this feature increases the prediction.

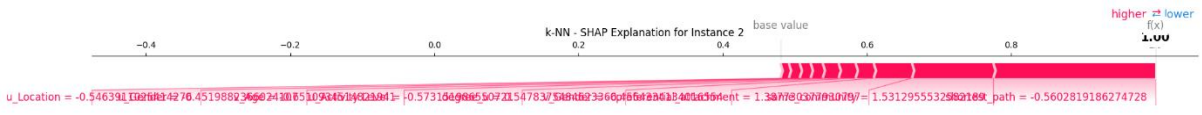


Figure 5.4.4: Contribution of the feature "Location" and "Activity Level" In the model

Figure 5.4.4 depicts the contribution of "Location" and "Activity Level." "Location" contributes negatively to the model's prediction shown in red indicating that higher values of "Location" decrease the predicted outcome. "Activity Level," on the other hand, contributes positively represented in blue suggesting that higher values of "Activity Level" lead to an increased predicted outcome.

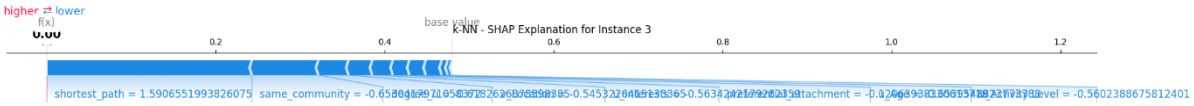


Figure 5.4.5: Contribution of the feature "Shortest Path" and "Same Community" In the model

Figure 5.4.5 presents the contributions of "Shortest Path" and "Same Community." "Shortest Path" contributes positively to the predicted outcome shown in blue indicating that higher values of this feature lead to a favorable prediction. "Same Community," however, has a larger negative effect shown in red suggesting that higher values of "Same Community" reduce the predicted outcome more significantly.

As shown in Figures 5.4.6, 5.4.7 and 5.4.8 SHAP (SHapley Additive exPlanations) contribution plots provide a graphical representation of the significance of individual features within a machine learning model. These plots demonstrate the contribution of different features such as "Shortest Path," "Same Community," and "Preferential Attachment" to the models predictions for specific instances.

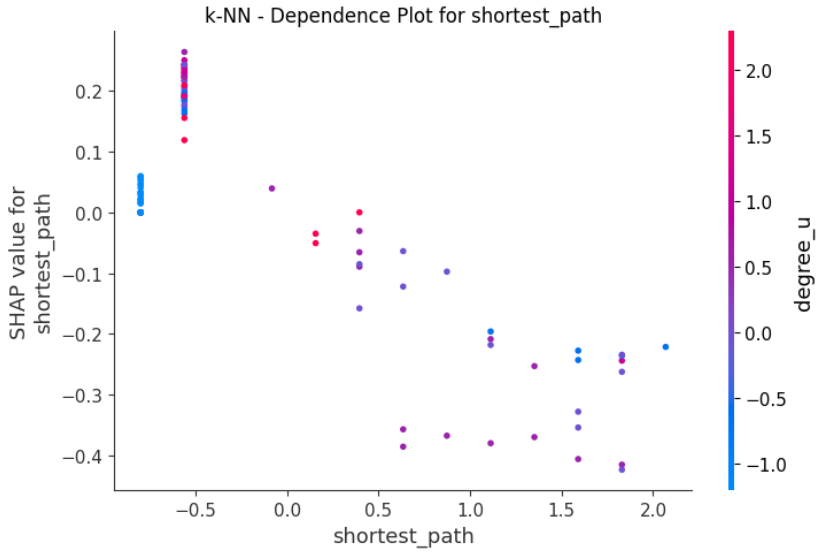


Figure 5.4.6: Contribution of the feature " shortest path " In the model

Figure 5.4.6 illustrates the SHAP contribution for the feature "Shortest Path." The plot shows a mix of positive (blue) and negative (red) contributions. As the value of "Shortest Path" increases the SHAP value increases indicating a positive contribution to the model's prediction. Conversely, lower values of "Shortest Path" result in a negative contribution, suggesting that this feature has a dual role depending on its value.

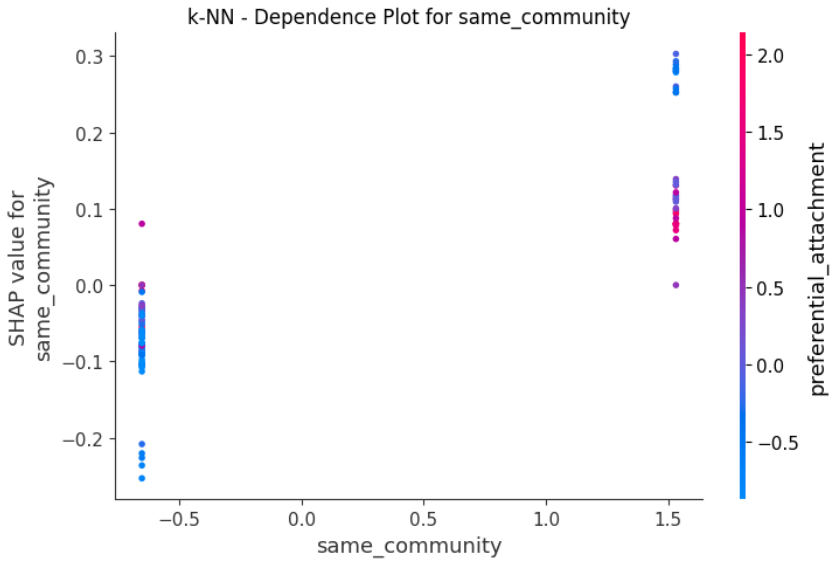


Figure 5.4.7: Contribution of the feature "Same Community" In the model

Figure 5.4.7 depicts the SHAP contribution for the feature "Same Community". The plot reveals a negative contribution where higher values of "Same Community" lead to a decrease in the predicted probability indicated by the red color. This suggests that "Same Community" negatively affects the prediction with higher values reducing the likelihood of the predicted outcome.

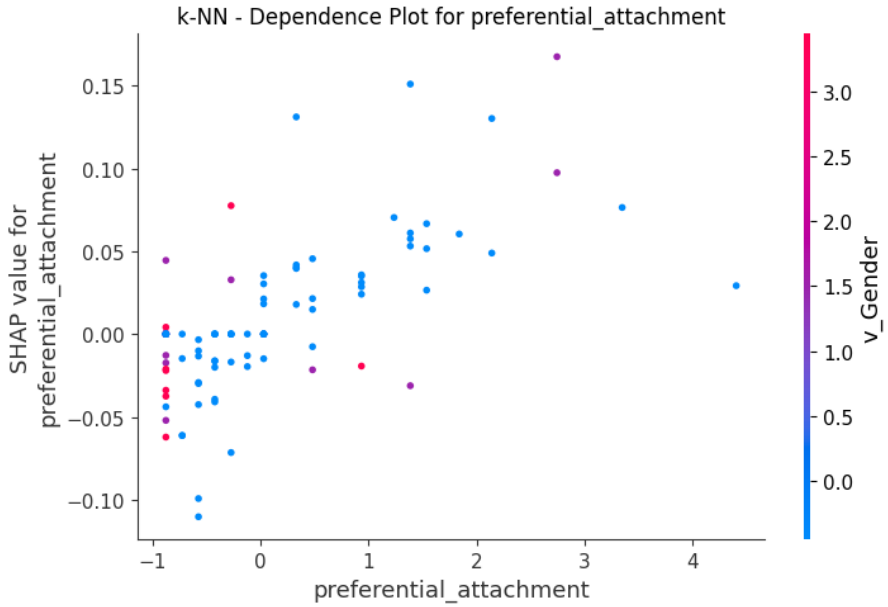


Figure 5.4.8: Contribution of the feature " Preferential Attachment " In the model

Figure 5.4.8 presents the SHAP contribution for the feature "Preferential Attachment". The plot shows that higher values of "Preferential Attachment" contribute positively (blue) while lower values have a less pronounced effect on the predicted outcome. The blue color indicates that as the value of "Preferential Attachment" increases the predicted outcome tends to increase as well.

5.5 XAI (Explainable AI) Results Comparison Summary

In the comparison of the XAI (Explainable AI) results for the Keras DNN and k-NN models both models reveal important insights into feature contributions but differ in their focus and impact. For the Keras DNN model global explainability reveals that features like shortest_path and same_community have the highest contributions to the predictions. These features are integral to the model's decision-making, influencing outcomes substantially. Features related to user interests such as u_Interests and v_Interests also show significant importance. The feature importance graph and SHAP analysis emphasize the consistent effect of these features across different instances. However user-specific features like v_Gender and v_Age have a relatively smaller role. Local explainability further details how features such as shortest_path and same_community influence individual predictions providing a dual-role impact depending on their values.

In contrast the k-NN model also identifies shortest_path and same_community as the most impactful features with a strong correlation to the target variable. Like the Keras DNN model features like u_Interests and v_Interests demonstrate notable importance in predicting outcomes. The global SHAP values show that the influence of these features is consistent across various instances. However, for k-NN features like preferential_attachment and v_Activity Level have a slightly more substantial impact compared to Keras DNN. In local explainability the k-NN model's SHAP plots highlight how specific features including Location and Activity Level contribute positively or negatively to predictions showcasing a more nuanced impact in certain cases.

Both models underscore the significance of network-based features especially shortest_path and same_community but they differ in the weight given to user activity and interest features. The Keras DNN model places slightly less emphasis on user-related attributes while the k-NN model incorporates these more prominently. Additionally, both models show how local feature impacts vary providing insights into model behavior for individual predictions.

Chapter 6: Conclusions

6.1 Conclusions

This research comprehensively explored the use of machine learning techniques including traditional algorithms and deep learning models for Predicting Future User Connections in social networks. By integrating both topological features such as node degree, common neighbors and non-topological features including user behavior, preferences and interaction history. The study developed a robust framework for predicting future connections. The use of classical machine learning algorithms like Random Forest, Support Vector Machines and k-Nearest Neighbors demonstrated effective baseline performance. However, deep learning models particularly deep neural networks proved superior in capturing complex hierarchical relationships within large and dynamic datasets. These models excelled at learning intricate patterns and nonlinearities that traditional models might overlook.

The ensemble methods further improved prediction accuracy by combining the strengths of multiple algorithms reducing bias and variance. The results highlight the significant impact of incorporating behavioral and preference-based features alongside network structure for more personalized and accurate Predicting Future User Connections . This approach is highly relevant for practical applications, such as enhancing friend recommendations on social media platforms, identifying potential security threats by forecasting suspicious link formation and improving product or content recommendations in e-commerce systems.

Overall, this research validates that a hybrid methodology combining diverse machine learning models and rich feature sets can substantially improve the quality and reliability of Predicting Future User Connections tasks. It also emphasizes the promising role of deep learning to address challenges posed by the complexity and scale of modern social networks. These findings contribute valuable insights to the evolving field of social network analysis and provide a strong foundation for future innovations in predictive modeling and network science.

6.2 Limitations & Recommendations

While the results are encouraging the study has some limitations. The dataset size though comprehensive is limited to publicly available social network data and may not fully capture all real-world complexities. The model performance could be affected by missing or noisy user behavior data. Additionally, dynamic changes in social networks over time were not fully incorporated which could influence link formation patterns. It is recommended to expand the dataset with more diverse and time-series data and to explore dynamic models that can adapt to evolving networks. Moreover, ethical considerations should be kept in mind when applying Predicting Future User Connections to ensure user privacy and data security.

6.3 Future Work

Future research will focus on refining deep learning architectures such as recurrent neural networks (RNNs) and graph neural networks (GNNs) to better model temporal and structural dynamics of social networks. Exploring hybrid models that combine traditional machine learning with deep learning could leverage the strengths of both approaches. Incorporating temporal features and real-time data processing will enhance adaptability to evolving social interactions. Additionally, applying explainable AI methods will improve interpretability and trustworthiness of deep learning models. Expanding applications to other domains like fraud detection, communication networks and recommendation engines will further validate and extend the impact of this work.

References

R. N. Lichtenwalter, J. T. Lussier, and N. V. Chawla, “New perspectives and methods in link prediction,” in Proc. 16th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, Washington, DC, USA, Jul. 2010, pp. 243–252. [Online]. Available: https://doi.org/10.1145/1835804.1835837

Z. Liu, Q.-M. Zhang, L. Lü, and T. Zhou, “Link Prediction in Complex Networks: A Local Naïve Bayes Model,” EPL (Europhysics Letters), vol. 96, no. 4, p. 48007, 2011. [Online]. Available: https://arxiv.org/pdf/1105.4005.

Anuradha Annaswamy, Henry Samueli & Boxue Du (2014). Comparative study of machine learning algorithms for link prediction in social networks. Proceedings of the 2014 International Conference on Data Mining and Knowledge Management.https://ieeexplore.ieee.org/

M. Zhang and Y. Chen, "Link Prediction Based on Graph Neural Networks," in Advances in Neural Information Processing Systems (NeurIPS), vol. 31, Montréal, Canada, Dec. 2018. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2018/file/53f0d7c537d99b3824f0f99d62ea2428-Paper.pdf

M. G. Raeini, "Link Prediction Using Supervised Machine Learning based on Aggregated and Topological Features," arXiv preprint arXiv:2006.16327, Jun. 2020. [Online]. Available: https://arxiv.org/abs/2006.16327

N. N. Daud, S. H. A. Hamid, M. Saadoon, F. Sahran, and N. B. Anuar, "Applications of link prediction in social networks: A review," Journal of Network and Computer Applications, vol. 166,p.102716,Sep.2020.[Online].Available: https://www.sciencedirect.com/science/article/abs/pii/S1084804520301909

X. Xian, T. Wu, X. Ma, S. Qiao, Y. Shao, C. Wang, L. Yuan, and Y. Wu, "Generative Graph Neural Networks for Link Prediction," arXiv preprint* arXiv:2301.00169, Jan. 2022. [Online]. Available: https://arxiv.org/abs/2301.00169

W. Li, T. Li, and K. Berahmand, "An effective link prediction method in multiplex social networks using local random walk towards dependable pathways," Journal of Combinatorial Optimization, vol. 45, no.1,pp.1–27,Jan.2023.[Online].Available: https://doi.org/10.1007/s10878-022-00961-z

M. Qin and D.-Y. Yeung, "Temporal Link Prediction: A Unified Framework, Taxonomy, and Review," ACM Computing Surveys, vol. 56, no. 4, Art. 89, Nov. 2023,doi: 10.1145/3625820. [Online]. Available: https://dl.acm.org/doi/10.1145/3625820

S. Khan, M. Badiy, F. Amounas, and M. Azrou, “A Powerful Approach to Improve Link Prediction Accuracy in Directed Social Networks Based on Ensemble Learning Models and Advanced Feature Extraction Techniques,” unpublished, 2024. [Online]. Available: https://www.researchgate.net/publication/383650831

X. Huang, J. Li, and Y. Yuan, “Link Prediction in Dynamic Social Networks Combining Entropy, Causality, and a Graph Convolutional Network Model,” vol. 26, no. 6, art. no. 477, May 2024. [Online]. Available: https://doi.org/10.3390/e26060477

M. Sun and M. Tang, “A Review of Link Prediction Algorithms in Dynamic Networks,” Mathematics, vol. 13, no. 5, art. 807, Feb. 2025, doi:10.3390/math13050807. [Online]. Available: https://www.mdpi.com/2227-7390/13/5/807

Formula References

1. L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001. DOI: 10.1023/A:1010933404324.

2. C. Cortes and V. Vapnik, "Support-Vector Networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995. DOI: 10.1007/BF00994018.

3. T. Cover and P. Hart, "Nearest Neighbor Pattern Classification," IEEE Trans. Inf. Theory, vol. 13, no. 1, pp. 21–27, 1967. DOI: 10.1109/TIT.1967.1053964.

4. D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning Representations by Backpropagating Errors," Nature, vol. 323, pp. 533–536, 1986. DOI: 10.1038/323533a0.

5. T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," Proc. 22nd ACM SIGKDD, 2016. arXiv:1603.02754.

6. F. Chollet, Keras Documentation, 2015. [Online]. Available: https://keras.io.

7. A Vaswani et al., "Attention Is All You Need," Adv. Neural Inf. Process. Syst. (NeurIPS), 2017. arXiv:1706.03762.

8. T. G. Dietterich, "Ensemble Methods in Machine Learning," Proc. First Int. Workshop MCS, 2000. DOI: 10.1007/3-540-45014-9_1.

9. D. H. Wolpert, "Stacked Generalization," Neural Networks, vol. 5, no. 2, pp. 241–259, 1992. DOI: 10.1016/S0893-6080(05)80023-1.

10. L. Breiman, "Bagging Predictors," Machine Learning, vol. 24, no. 2, pp. 123–140, 1996. DOI: 10.1007/BF00058655.

11. Y. Freund and R. E. Schapire, "A Decision-Theoretic Generalization of On-Line Learning," J. Comput. Syst. Sci., vol. 55, no. 1, pp. 119–139, 1997. DOI: 10.1006/jcss.1997.1504.

12. J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," Ann. Stat., vol. 29, no. 5, pp. 1189–1232, 2001. DOI: 10.1214/aos/1013203451.